ORF363/COS323 Convexity & Convex Optimization Part 3 (Lec6.pdf)

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Topics & Logistics

- Operations that preserve convexity
- Applications of Convex Optimization
  - in Statistics
  - in Machine Learning

The Midterm exam will be on Thursday March 5, 2:55 – 4:15 pm @ Robertson 016.

It will be in-person exam.

More information will be released shortly.

Theses slides will be covered in the next 2-3 lectures

Convexity preserving operations

- **<u>Operation 1:</u>** The nonnegative weighted sum of convex functions is a convex function. That is: if $f_1, f_2, \dots, f_m: R^n \to R$, convex functions, and $w_1, w_2, \dots, w_m \geq 0$ then
  $(\Rightarrow)$ $f(x) = w_1 f_1(x) + w_2 f_2(x) + \cdots + w_m f_m(x)$ is convex function

- **Proof**: in the pdf notes using the definition of convex function. But it can also be proven using the 2nd order characterization of convexity using the $\nabla^2 f_i(x)$ matrices (TODO as exercise!).

- **Exercises**: if f_1, f_2 are convex functions
  - (a) is $f_1 - f_2$ convex? (b) Is $f_1 f_2$ convex? (c) Is f_1 / f_2 convex?

Convexity preserving operations

- **<u>Operation 2:</u>** Composition of a convex function with an affine function.

  Suppose $f: R^n \to R$ and $g: R^m \to R$ with $g(x) = f(Ax + b)$ and $A \in R^{n \times m}$, and $dom(g) = \{x \in R^m \mid Ax + b \in dom(f)\}$. If f is convex function, then g is also a convex function.

- **Proof**: in the pdf notes and on blackboard.

- **Example**: is $f(x, y) = (x - 0.3y)^2 + e^{(0.5x - 0.8y)}$ convex?

Convexity preserving operations

- **<u>Operation 3:</u>** Pointwise maximum of convex functions. if $f_1, \dots, f_m: R^n \to R$ are convex functions then ($\Rightarrow$) $f(x) = \max\{f_1(x), \dots, f_m(x)\}$ is a convex function with $dom(f) = dom(f_1) \cap \cdots \cap dom(f)$.

- **Proof**: in the pdf notes and on blackboard.

- **Example**:

  - $f_1(x) = (x-2)^2$ and $f_2(x) = x^2$

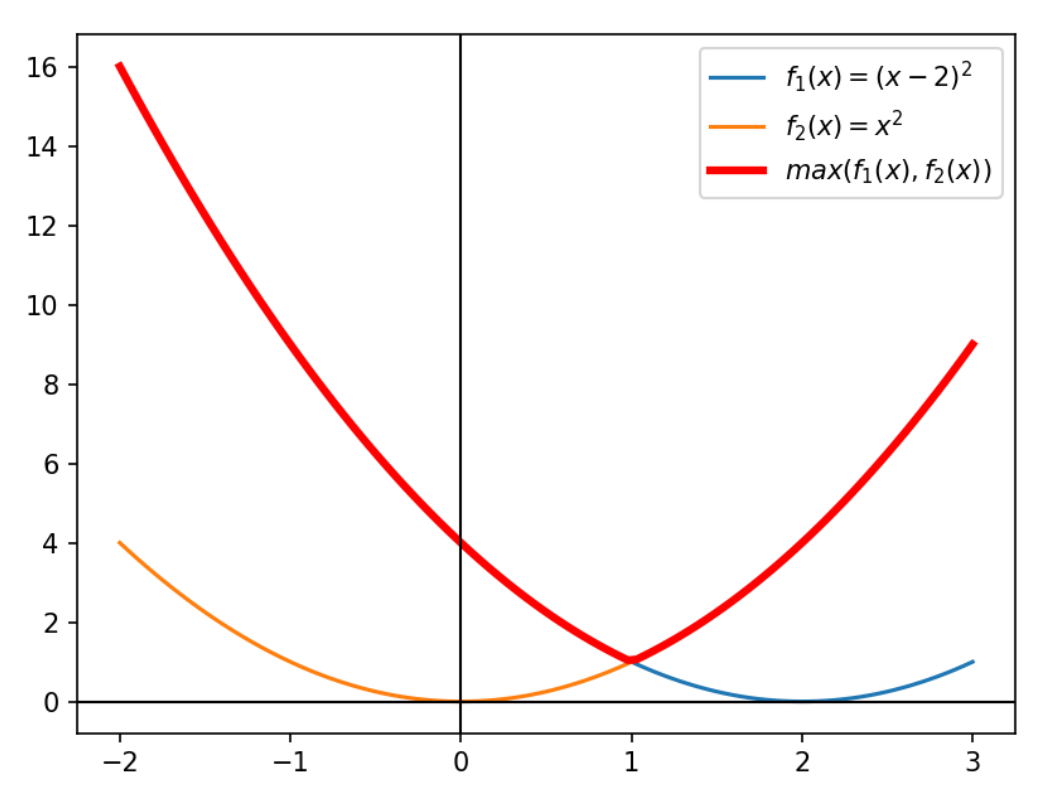
  - Find $f(x) = max\{f_1(x), f_2(x)\}$

Convexity preserving operations

- **Operation 3:** Pointwise maximum of convex functions. if $f_1, \dots, f_m : R^n \to R$ are convex functions then ($\Rightarrow$) $f(x) = \max\{f_1, \dots, f_m\}$ is a convex function with $dom(f) = dom(f_1) \cap \cdots \cap dom(f)$.

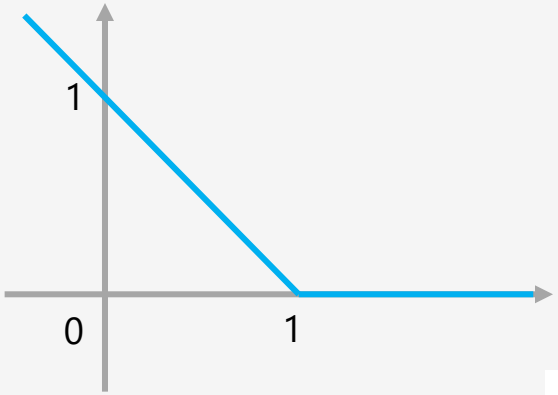
- **Proof**: in the pdf notes and on blackboard.

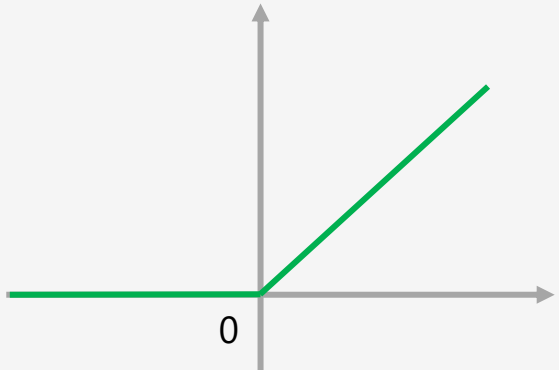
- **Example**:
  - $f_1(x) = (x-2)^2$ and $f_2(x) = x^2$
  - Find $f(x) = max\{f_1(x), f_2(x)\}$

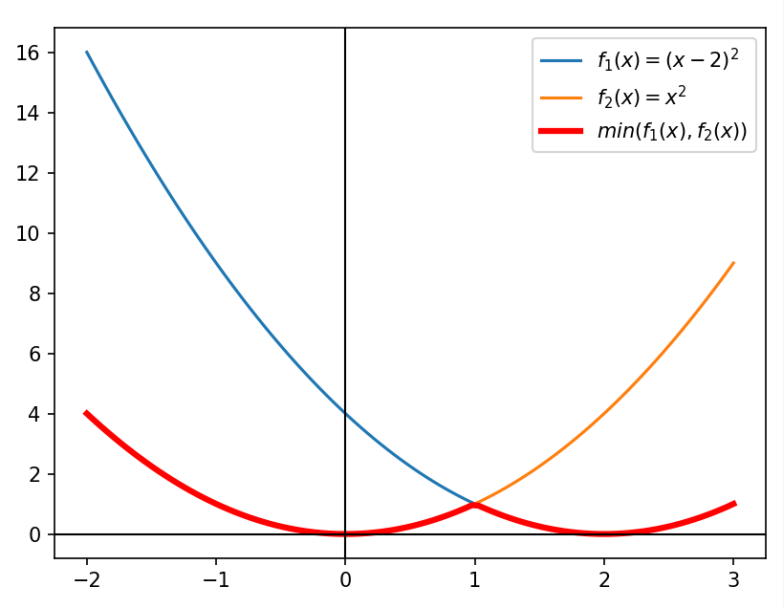


Convexity preserving operations

- **Operation 3:** Pointwise maximum of convex functions.
- **More examples**:
  - Hinge loss: $\max\{0, 1 - x\}$
  - ReLU: $\max\{0, x\}$
- **Remark:** the pointwise **minimum** of convex functions is **NOT** convex function







Convexity preserving operations

- **Operation 4**: Restriction to a line. If $f: R^n \to R$ is a convex function then ($\Rightarrow$) the function $g: R \to R$ with $g(\alpha) = f(x + \alpha y)$, for some fixed $x, y \in R^n$, is a convex function *(**notes**: α is not fixed, it varies in R; think of y as a direction).*

- **Proof**: in the pdf notes and on blackboard.

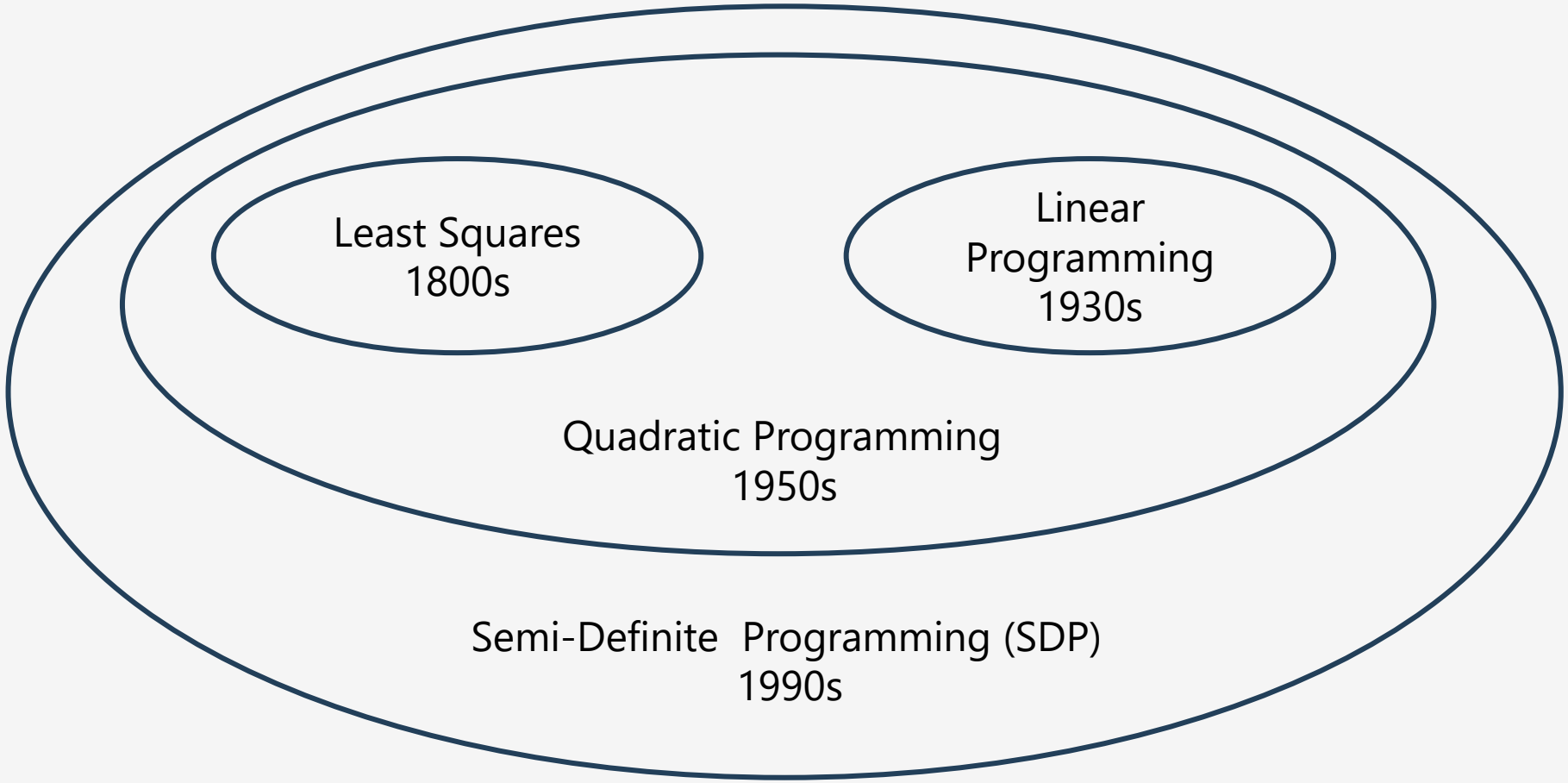
- **Remark**: this operation will be frequently used by the algorithms we will examine in future lectures. These algorithms minimize a convex function over a line (called ***line-search algorithms***) to determine the next approximation of the optimal solution they search for.

Types of optimization problems

Types of Optimization problems

- **Convex -vs- Nonconvex Optimization problems**
  - Convex functions have very favorable properties
  - Nonconvex functions have many local optima
- **Continuous -vs- Discrete (Integer) Optimization**
  - Main difference: the values of the decision vars
  - Integer Optimization is computationally expensive
- **Deterministic -vs- Stochastic/Robust Optimization**
  - Deterministic → all data are known precisely
  - Stochastic → data may be random or uncertain
- **Single-objective -vs- Multi-objective Optimization**
  - When we have only one or many objectives to optimize
  - The constraint set is always the same
- **Linear Optimization**
  - Objective linear
  - Constraints affine
- **Nonlinear Optimization**
  - At least one constraint is nonlinear or the objective is nonlinear
- **Local Optimization**
  - Happy with finding only one local optimum
- **Global Optimization**
  - Only happy with finding the best local optimum

Convex Optimization Hierarchy (for this course)



Least Squares

Least squares – Personalized pricing of flight tickets

THE STORY

- You have been hired by Priceline.com
- You will be working on the **"Name your own price" tool**
  - Customers bid on the price they want to pay for a flight ticket
  - IF the **bid is high enough** THEN **the company sells the ticket** to the customer
  - ELSE the **bid is rejected**, and the company makes a **counteroffer** (a higher price)
- Your job is to optimize the pricing strategy
  - i.e., predict the highest price a future customer is willing to pay for a flight

Personalized pricing of flight tickets – The data

- For each **customer data** representing different **attributes/features** are available:
  - Customer age
  - # of Facebook friends
  - Residence address
  - Travel frequency
  - Gender
  - Marital status
  - Their favorite movie
  - …

- For each customer we also know:
  - The price they paid for the ticket *(aka, **label**)*

How can I formulate my data in vectors and matrices?

- For each customer different attribute/feature values are available: → consider the i-th customer
  - Customer age → $x_1^i = 52$
  - # of Facebook friends → $x_2^i = 0$
  - Residence address → $x_3^i = 314\ XYZ\ Street, \mathrm{NJ}\ 08540$
  - Travel frequency → $x_4^i = 2$
  - Gender → $x_5^i = M$
  - Marital status → $x_6^i = M$
  - Their favorite movie → $x_7^i = Cinema\ Paradiso$
  - … up to p-th attribute → $x_p^i = \cdots$

- For each customer we also know:
  - The price they paid for the ticket → $y^i = \$485$ *(aka, **label**)*

How can I formulate my data in vectors and matrices?

- For each customer different attribute/feature values are available: → consider the i+1 customer
  - Customer age → $x_1^{i+1} = 28$
  - # of Facebook friends → $x_2^{i+1} = 500$
  - Residence address → $x_3^{i+1} = 2718\ ABC\ Street, \mathrm{NC}\ 27513$
  - Travel frequency → $x_4^{i+1} = 4$
  - Gender → $x_5^{i+1} = F$
  - Marital status → $x_6^{i+1} = S$
  - Their favorite movie → $x_7^{i+1} = Casablanca$
  - … up to p-th attribute → $x_p^{i+1} = \cdots$

- For each customer we also know:
  - The price they paid for the ticket → $y^i = \$475$ *(aka, **label**)*

How can I formulate my data in vectors and matrices?

- For each customer different attributes/features are available: → collect all attribute data for all customers
  - Customer age → $\boldsymbol{x_1} = (x_1^1, x_1^2, \dots, x_1^n)^T$
  - # of Facebook friends → $\boldsymbol{x_2} = (x_2^1, x_2^2, \dots, x_2^n)^T$
  - Residence address → $\boldsymbol{x_3} = (x_3^1, x_3^2, \dots, x_3^n)^T$
  - Travel frequency → $\boldsymbol{x_4} = (x_4^1, x_4^2, \dots, x_4^n)^T$
  - Gender → $\boldsymbol{x_5} = \left(x_5^1, x_5^2, \dots, x_5^n\right)^T$
  - Marital status → $\boldsymbol{x_6} = (x_6^1, x_6^2, \dots, x_6^n)^T$
  - Their favorite movie → $\boldsymbol{x_7} = (x_7^1, x_7^2, \dots, x_7^n)^T$
  - … up to p-th attribute→ $\boldsymbol{x_p} = \left(x_p^1, x_p^2, \dots, x_p^n\right)^T$

The total number of customers is: $\boldsymbol{n}$

$\boldsymbol{x_i} \in R^n$ and $\boldsymbol{y} \in R^n$

- For each customer we also know:
  - The price they paid for the ticket → $\boldsymbol{y} = (y^1, y^2, \dots, y^n)^T$

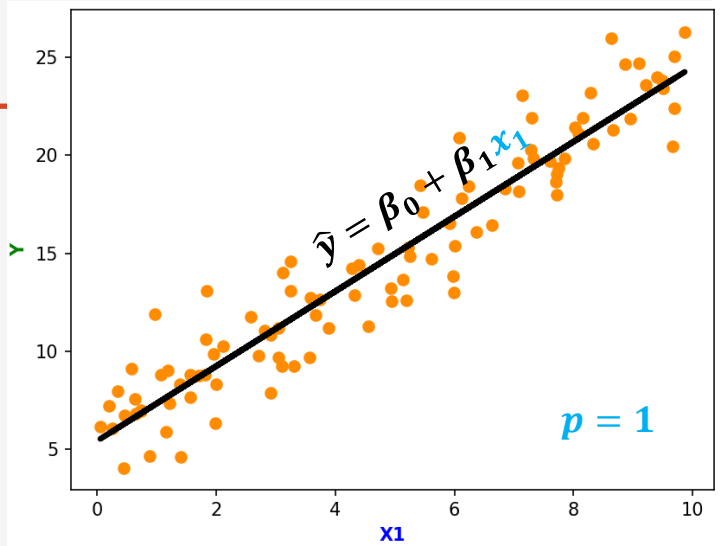
    *(also known as, **label**)*

Our **dataset**: $\boldsymbol{D} = \left\{ z^j = \left(\left(x_1^j, x_2^j, \dots, x_p^j \right), y^j \right) \middle| \; j = 1,2, \dots, \boldsymbol{n} \right\}$

Learning the relation between features and labels

- Note that we have data for the attributes (features) and the labels
- We are interested in **learning a function $\boldsymbol{f}: \boldsymbol{\mathcal{X}} \rightarrow \boldsymbol{\mathcal{Y}}$ (or model)** that takes as **input** an attribute vector $\boldsymbol{x^j} \in \boldsymbol{\mathcal{X}} \subseteq R^p$ and **outputs** the corresponding label $y^j \in \boldsymbol{\mathcal{Y}} \subseteq R$ (the price the customer was willing to pay)
  - Note: $\boldsymbol{x^j}$ represents the data collected by the company for the ***j*-th customer** and looks like:
    - $\boldsymbol{x^j} = \left(x_1^j, x_2^j, \dots, x_p^j\right)^T \in \boldsymbol{\mathcal{X}} \subseteq R^p$ for $j = 1,2, \dots, n$
    - We have data for $\boldsymbol{n}$ **customers**.
- Once we *learn* the function $\boldsymbol{f}$ we can use it to ***predict*** the new label (prices) for a new customer by inputting her/his attributes → $y^{new} = f(\boldsymbol{x^{new}})$

Decide on the fitting model



- There are many functions that **relate the labels to the attributes**
- A simple **linear regression** will look like this:

$$\boxed{f(\boldsymbol{\beta}) = \hat{y}^j = \beta_0 + \beta_1 x_1^j + \beta_2 x_2^j + \cdots + \beta_p x_p^j}, \quad \text{for all } j = 1,2,\ldots,n$$

- We are looking for the best (optimal) values of the unknown parameters: $\beta_0, \beta_1, \ldots, \beta_p$
  - **Note**: $\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_p)^T \in R^p$ are our **decision variables**
- In vector form we can rewrite the above equation

$$\boxed{\hat{\boldsymbol{y}} = \beta_0 \mathbf{1} + \beta_1 \boldsymbol{x_1} + \beta_2 \boldsymbol{x_2} + \cdots + \beta_p \boldsymbol{x_p}} \Leftrightarrow \boxed{\hat{\boldsymbol{y}} - \beta_0 \mathbf{1} - \beta_1 \boldsymbol{x_1} - \beta_2 \boldsymbol{x_2} - \cdots - \beta_p \boldsymbol{x_p} = 0}$$

where $\boldsymbol{x_i} \in \boldsymbol{R^n}$ for $i = 1,2,\ldots,p$, and $\hat{\boldsymbol{y}} \in \boldsymbol{R^n}$ is the **<u>estimate</u>** of the true label (price) $\boldsymbol{y} \in \boldsymbol{R^n}$

Formulate the optimization problem to compute the βs

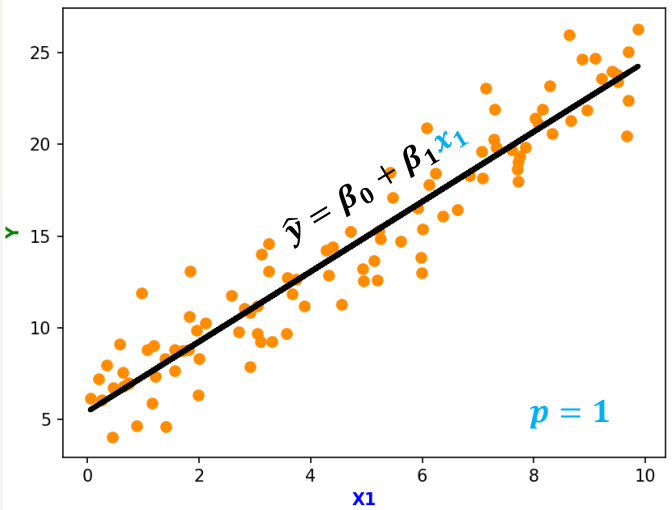
$$\overbrace{y - \hat{y}}^{\textbf{deviations}} \approx 0 \Leftrightarrow y - \overbrace{\beta_0 \mathbf{1} - \beta_1 x_1 - \beta_2 x_2 - \cdots - \beta_p x_p}^{\hat{y} = f(\beta)} \approx 0 \Leftrightarrow y - \overbrace{\beta_0 \mathbf{1} - \sum_{k=1}^{p} \beta_k x_k}^{\hat{y} = f(\beta)} \approx 0$$

- Remember **Least Squares?**
  - Use it to compute the optimal $\boldsymbol{\beta}$ so that the **deviation** $||y - \hat{y}||$ is **minimized**

$$min_{\beta \in R^{p+1}} ||y - \hat{y}|| = min_{\beta \in R^{p+1}} \left\| y - \beta_0 \mathbf{1} - \sum_{k=1}^{p} \beta_k x_k \right\|_2 = min_{\beta \in R^{p+1}} \left|\left| y - X\boldsymbol{\beta} \right|\right|_2$$

$$X = [\mathbf{1}, x_1, x_2, \ldots, x_p] \in R^{n \times (p+1)}, \quad \mathbf{1}, x_i \in R^n, \quad y \in R^n, \quad \longleftarrow \textbf{Data}$$

$$\boldsymbol{\beta} = (\beta_0, \beta_1, \ldots, \beta_p)^T \in R^{p+1} \quad \longleftarrow \textbf{Decision variables}$$



Discussion on the optimal values of $\boldsymbol{\beta}$ from LS

- **Observation:** all β_is will be $\neq 0$
  - The model treats all attributes (or features) are **equally relevant**
- But what if we have features that are **irrelevant** (i.e., favorite movie)?
  - If only there was a model that can assign optimal values to the **relevant** βs
- If we solve a modified optimization problem our wish will be granted ☺

Regularization term

$$min_{\beta \in R^p} \left\| \boldsymbol{y} - \beta_0 \mathbf{1} - \sum_{k=1}^{p} \beta_k \boldsymbol{x_k} \right\|_2 + \gamma \left\| \boldsymbol{\beta} \right\|_1 = min_{\beta \in R^p} \left\| \boldsymbol{y} - \boldsymbol{X}\boldsymbol{\beta} \right\|_2 + \gamma \left\| \boldsymbol{\beta} \right\|_1$$

LASSO model

- The magic parameter γ can be selected appropriately to **enforce $\boldsymbol{\beta}$s of irrelevant features → 0**

Selecting the right value of γ

$$min_{\beta \in R^p} \left\| y - \beta_0 1 - \sum_{k=1}^{p} \beta_k x_k \right\|_2 + \gamma \left| |\beta| \right|_1$$

- If γ is **large**, then more emphasis is placed on **minimizing the L_1 norm** (regularization term)
  - Irrelevant features will have their βs driven to 0
  - Our model will have smaller number of predictors → more generalizable
- If γ is **small**, then more effort is placed on **minimizing the L_2 norm**
  - Makes the prediction (deviation) error small
- **Finding the right γ** is done by trying many different values and selecting the model that performs best

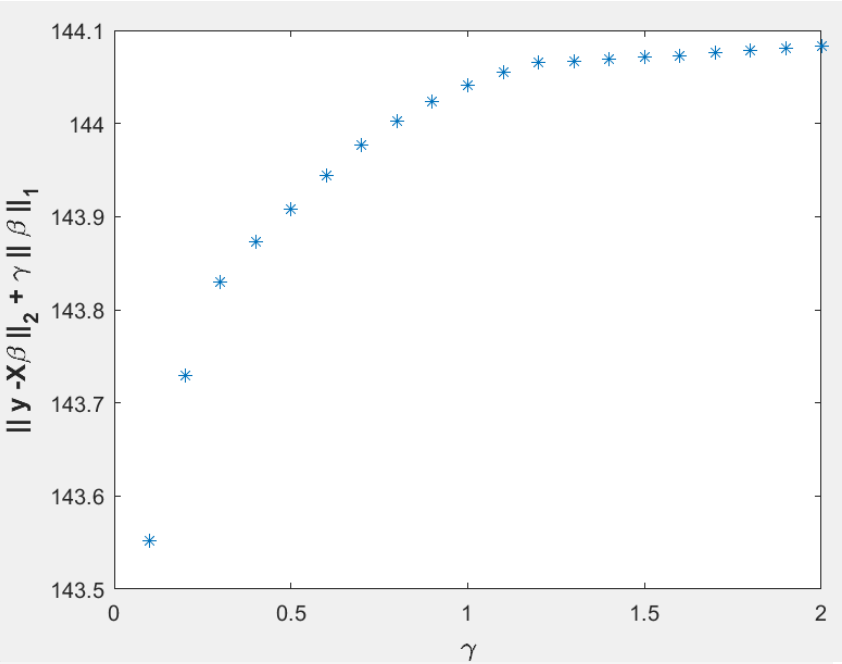
Solve a large LS problem with CVX & MATLAB

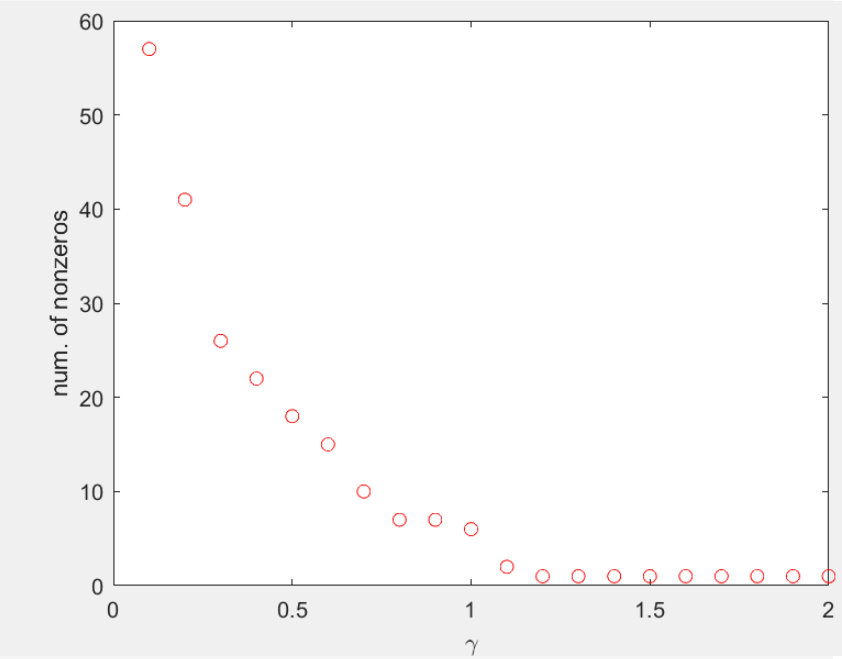
```
n = 10000;
p = 100;
error_vec = [];
n_nonzeros = [];
X = [ones(n,1),  rand(n, p)];
y = 5*rand(n,1);

for gamma = 0.1:0.1:2
    cvx_begin
        variable b(p+1)
        minimize(norm(y-X*b,2) + gamma*norm(b,1))
    cvx_end
    error_vec = [error_vec; norm(y-X*b,2)];
    n_nonzeros = [n_nonzeros; length(find(abs(b)>0.01))]
End

plot(0.1:0.1:2, error_vec, '*')
figure, plot(0.1:0.1:2, n_nonzeros, 'ro')
```

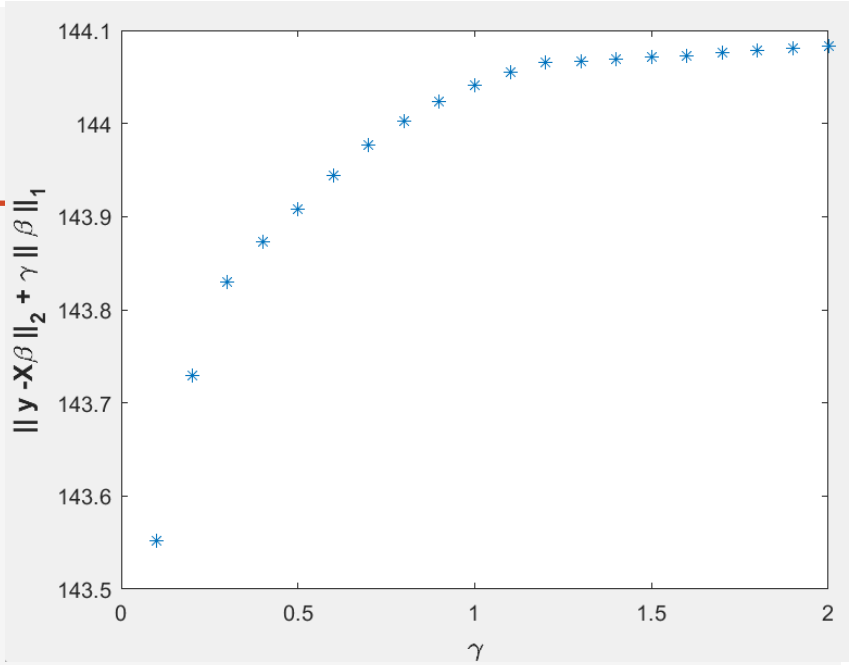
- `n = 10000;` → Number of data points
- `p = 100;` → Number of features/attributes
- `error_vec = [];` → Objective function of LS problem
- `n_nonzeros = [];` → # of nonzero β_is for $i = 0,1,\dots,p$
- `X = [ones(n,1), rand(n, p)]; y = 5*rand(n,1);` } Generate random data $X \in R^{n\times(p+1)}$, $y \in R^n$
- `for gamma = 0.1:0.1:2` → We generate 20 values or γ, and solve the corresponding 20 optimization problems **(in only 74 secs on my laptop)**
- `End` — For each $\gamma \in \{0.1, 0.2, \dots, 2\}$ we compute a different optimal solution $\beta \in R^p$. Each vector β is used to create a prediction model.

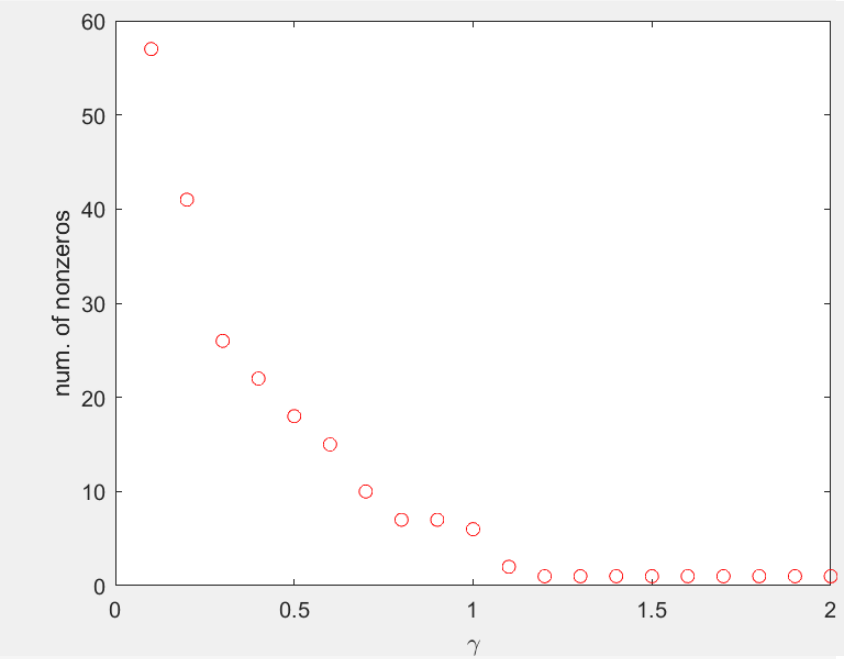




Solve a large LS problem with CVX (super fast)

- In the `for-loop` we consider **20 different values** for $\boldsymbol{\gamma} = \mathbf{0.1, 0.2, \ldots, 2}$
- We solve **20 optimization problems**, one for each γ value
- Each of these optimization problems has a different optimal solution (different $\beta \in R^p$)
- In a way we learnt **20 different Lease Squares models** – one for each optimal β vector
- If we store the different $\boldsymbol{\beta}$s <u>we can use any of these models to make predictions</u>.
- **As the value of γ increases**
  - **The objective of the optimization problem (error or deviations) increases** (see the blue asterisks)
  - **The number of nonzero elements in β decreases** (see the red circles)
  - <u>Even if we have a small number of nonzero elements in our β the error does not crease much compared to the case when β has many zeros.</u>





The case of collinearity

- **Question (from the Spring 2026 class):** What happens when there is collinearity in the data?
  - Collinearity is understood as when two (or more) features are highly correlated.
- **LASSO's behavior:**
  - It **picks at random one** of these features and **sets its coefficient to a nonzero**.
  - The other feature's coefficient is set to zero (i.e., it is not ignored)
  - **The result**: you still get a sparse model (many of the coefficients become zero)
  - BUT then you do not know if the one picked from the colinear features is the right one
- **Elastic net** (an extension of LASSO) is more appropriate model for such cases
  - The penalty term $\gamma_2 ||\beta||_2$ encourages corelated features to be **picked up together.**

$$\boxed{min_{\beta \in R^p} \left\| y - \beta_0 1 - \sum_{k=1}^{p} \beta_k x_k \right\|_2 + \gamma_1 ||\beta||_1 + \gamma_2 ||\beta||_2}$$

Many papers implement the Elastic net, showing very good and robust results

Other models worth using (not covered in this course)

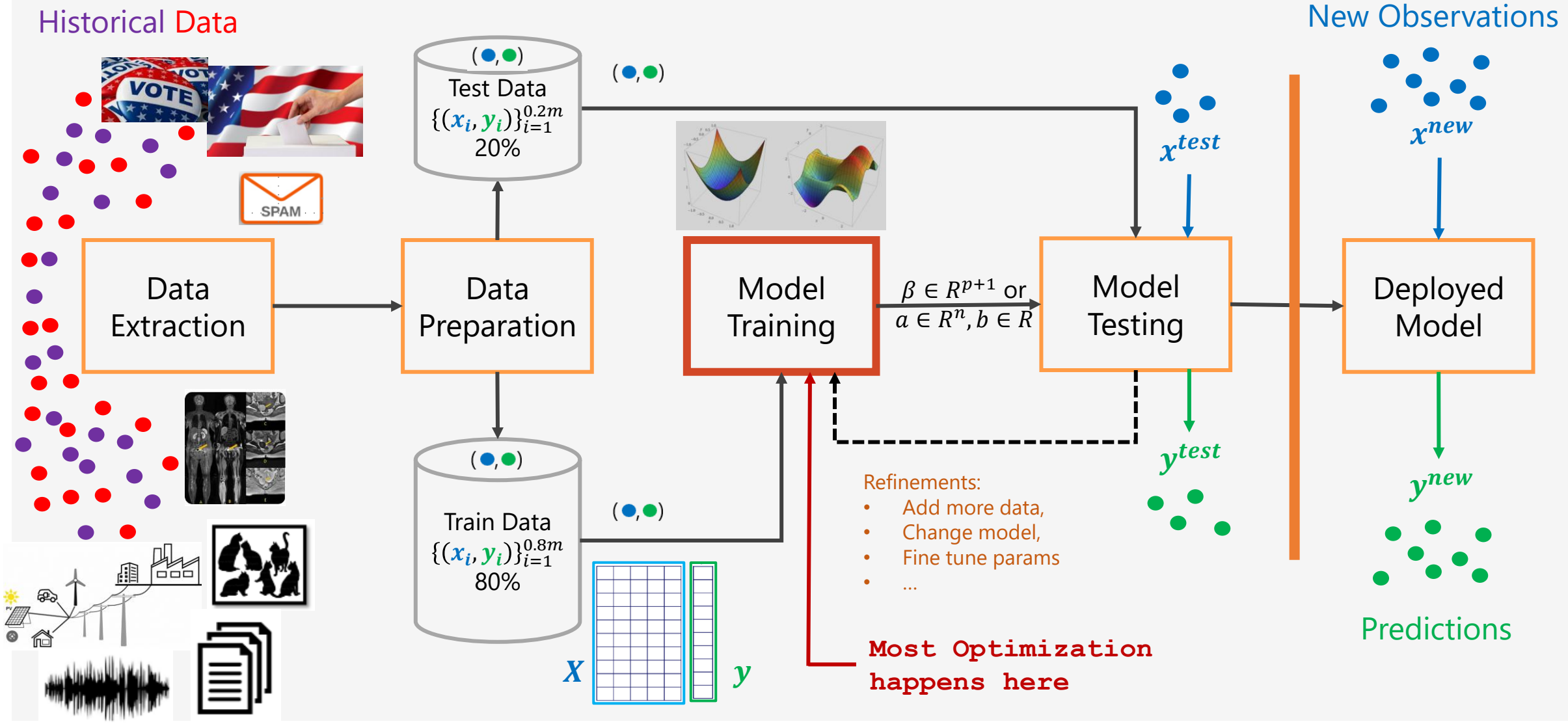
- **Question** What happens when your data have a natural group-structure (needed to be identified)?
  - This is still when your features contain highly correlated features (which may form groups)
- **Group-LASSO**

$$min_{\beta \in R^p} \left\| y - \beta_0 1 - \sum_{k=1}^{p} \beta_k x_k \right\|_2 + \gamma \sum_{g=1}^{G} \sqrt{p_g} \left\| \beta_g \right\|_2$$

  - β_g: the coefficient vector belonging to the g-th group of features
  - p_g: the number of features belonging to the g-th group of features
  - G: the number of groups the data can me split into

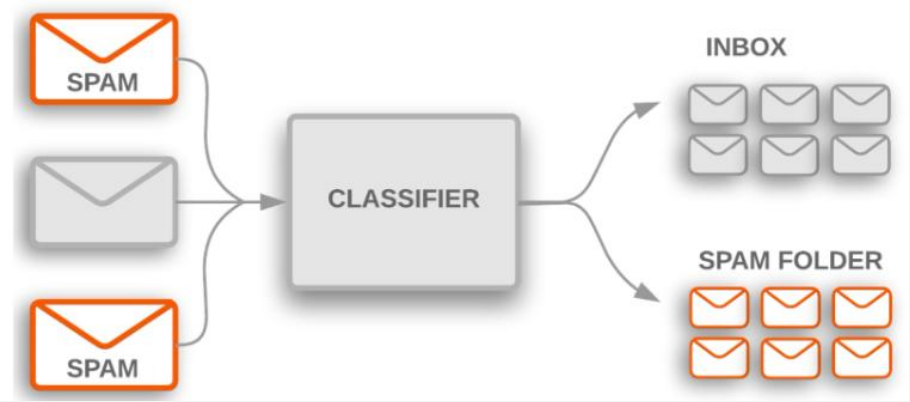
Support Vector Machines

Machine Learning pipeline for model computations



Support Vector Machines (SVM)

- **SVM**s are used for **supervised learning** tasks
- The problem of supervised learning is broadly defined as follows:
  - We are given a lot of historical data and their labels, and we want to **learn a model** that classifies new (unseen) data to their appropriate labels.
- **Example:** classify emails to *spam* or *not spam*
  - The emails are our data
  - The label is *"spam"* → *1* or *"not spam"* → *1*
  - We have many emails with their correct label → our **training set**
  - *We are looking for an algorithm and model that classifies future emails to spam or not spam*



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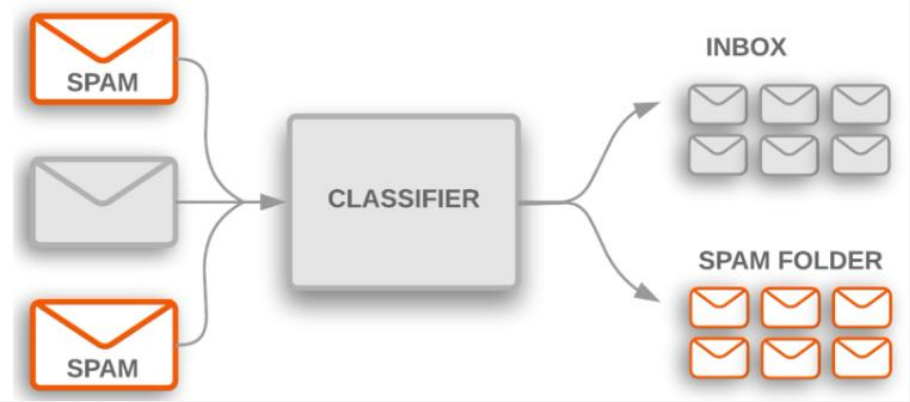

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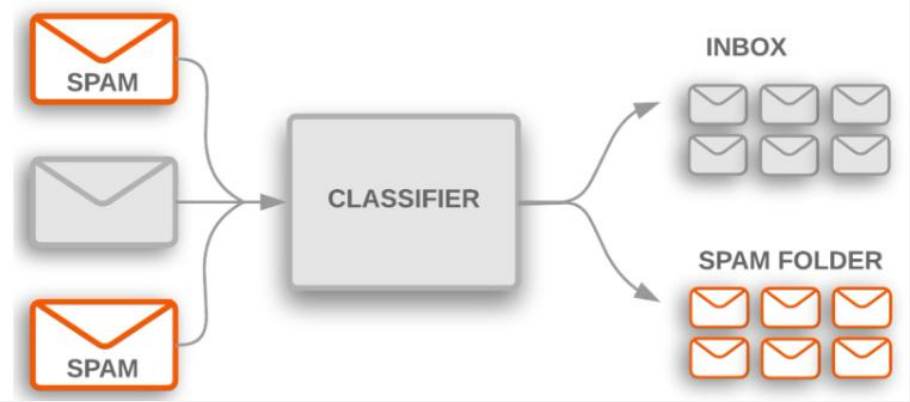
WHY DO YOU THINK IT IS CALLED SUPERVISED LEARNING??
AVAILABILITY OF LABELS



Support Vector Machines (SVM)

- **Examples of Binary Classification & Supervised Learning**
  - **Cancer Diagnosis** → Many MRI images with various cancers are collected and used to train models. These models are then used to assist doctors detect cancers earlier in new images and increase the survival rates of patients.
  - **Malware Classification** → very important to better understand how malware can infect computers or other devices, the threat level they pose and how to protect against them.
  - **Image or Text Sentiment Analysis** → use images or text as data to learn to identify a person's emotions about various issues (very useful in social media analytics)
  - **Anomaly detection** → very useful in recognizing strange patterns in financial transactions that could be fraudulent, or detect abnormal patterns in machines which may have catastrophic results (e.g., blackouts, car crashes, etc.)

Support Vector Machines (SVM)

- Examples of our data
  - **Email 1:** "Hello class, my office ours this week have moved to Thursday 4-5.30pm. Lecture 4 is now up on the course website. Thanks, Amirali." – Label "Not Spam"
  - **Email 2:** "good day, my name is bob. I seek a true love soulmate. Are you ready for relations? Check my profile: http://soul4.com/me.exe" – Label: "Spam"
  - **Email 3:** "Hey man, I am tired of this homework for ORF363. Let's go party tonight. I heard the new bar in Plainsboro is great and always crowded. We can always ask for an extension. –J." Label: "Spam"

Support Vector Machines (SVM)

- Our **features** are:
  - How many words are in the email?
  - How many misspelled words?
  - How many links?
  - Is there a $ sign?
  - Does the word “bank account” appear?
  - Is the sender’s email client trustworthy?
  - … up to the n-th feature

- Our **labels** are:
  - Spam (+1)
  - Not spam (-1)

Support Vector Machines (SVM)

- Our features are → and their names
  - How many words are in the email? → x^1
  - How many misspelled words? → x^2
  - How many links? → x^3
  - Is there a $ sign? → x^4
  - Does the word "bank account" appear? → x^5
  - Is the sender's email client trustworthy? → x^7
  - … up to the n-th feature → x^n

- Our label → y
  - $y \in \{-1, +1\}$
  - Spam (+1)
  - Not spam (-1)
  - ***Binary classification***

Support Vector Machines (SVM)

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  - Spam (+1)
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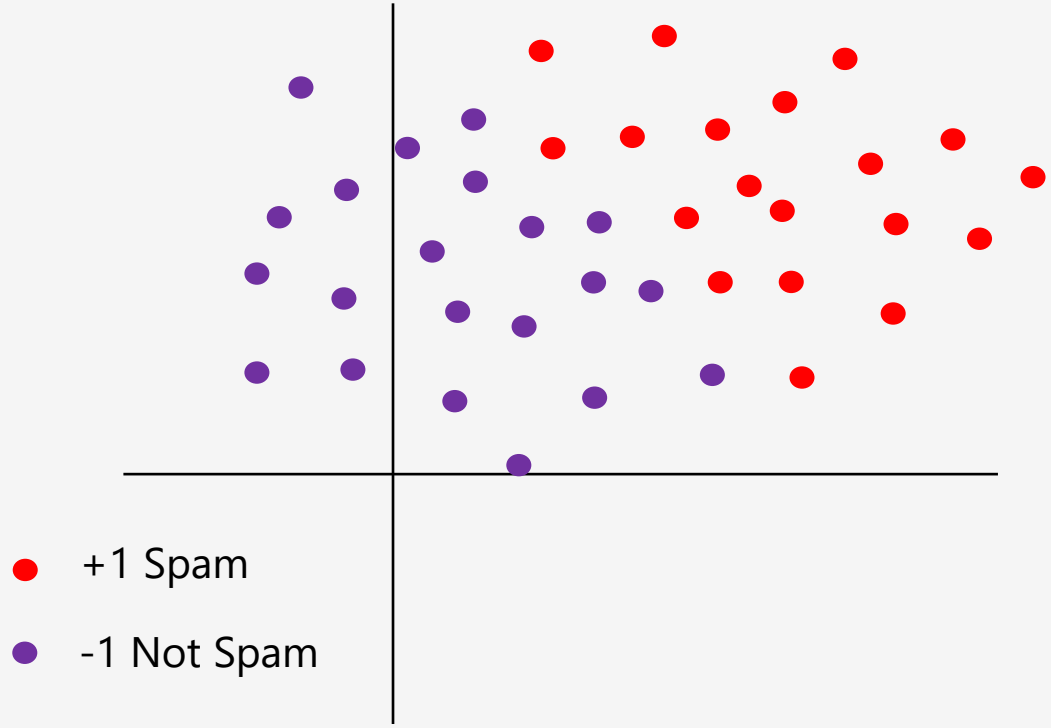
  - ***Binary classification***

- We have collected $\boldsymbol{m}$ **emails**.
- We stored in our ***training set:***

$$D = \{(x_i, y_i)\}_{i=1}^{m}$$

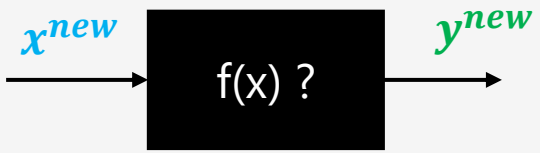
where $x_i = (x_i^1, x_i^2, \dots, x_i^n) \in R^n$ represents the i-th email.

Support Vector Machines (SVM)



<u>The Goal:</u>

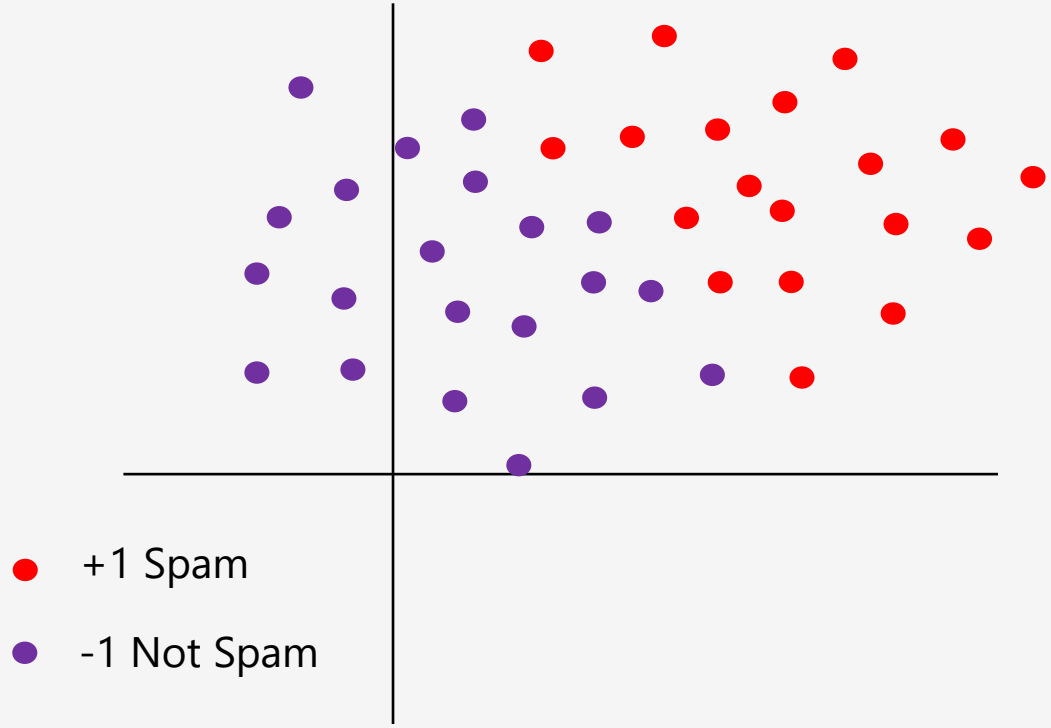
- Find a **classifier** $f: R^n \to \{-1, +1\}$ which can predict the label (spam or not spam) of a new email



If $f(x^{new}) > 0$, then $y^{new} = +1$
If $f(x^{new}) < 0$, then $y^{new} = -1$

Hence the equality $\boldsymbol{f(x) = 0}$ plays a crucial role!
What does it remind you??

Support Vector Machines (SVM)



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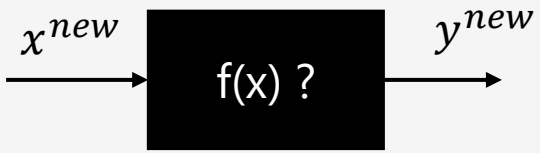

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The ZERO LEVEL SET of f

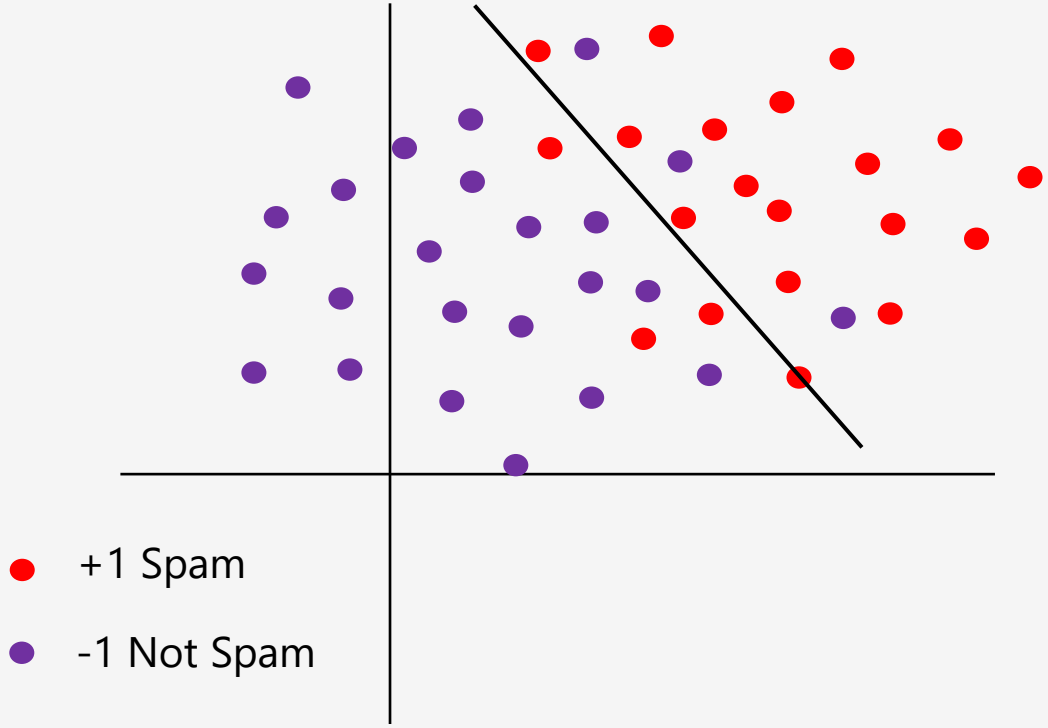
$$L_0 = \{x \in R^n \mid f(x) = 0\}$$

Support Vector Machines (SVM)

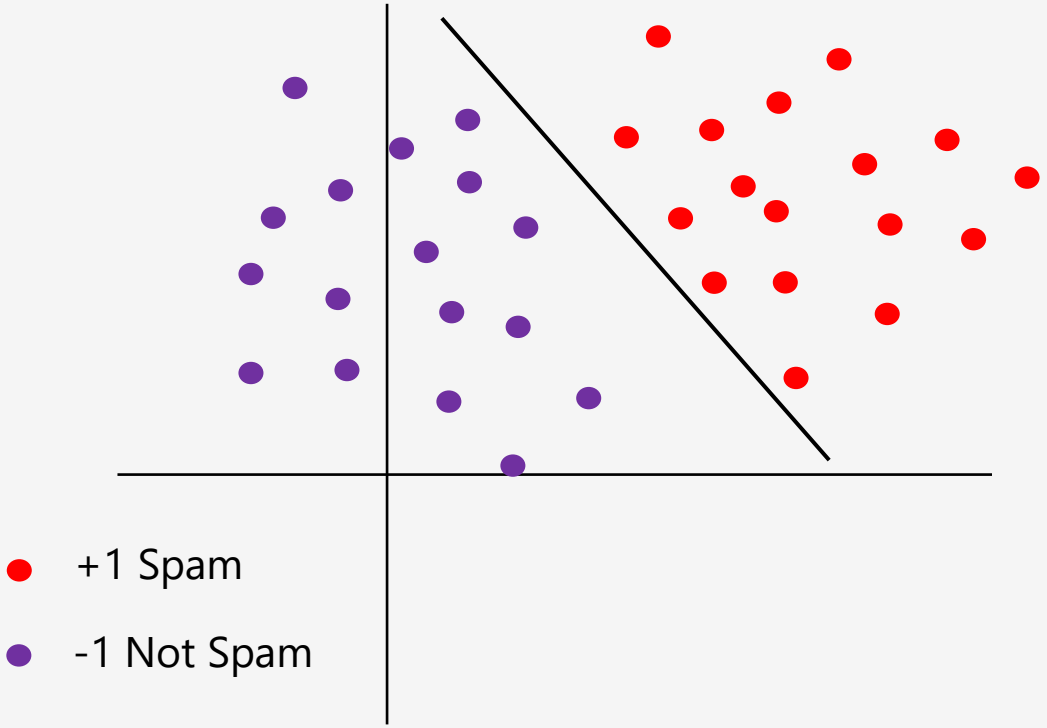
- **Finding the function f is not an easy problem**
  - There are so many functions *(e.g., linear, polynomial, general nonlinear, etc.)*
- SVMs consider only **affine functions**: $f(x) = a^T x - b$
  - Our unknows (**decision variables**) are $a \in R^n$ and $b \in R$
  - x is our feature vector (and it is known data)
- **Learning the classifier f** becomes computing the best $a \in R^n$ and b, such that:

$$\text{If} \quad a^T x_i - b \geq 0, \quad \text{then} \quad y_i = +1, \forall i$$
$$\text{If} \quad a^T x_i - b \leq 0, \quad \text{then} \quad y_i = -1, \forall i$$

Two cases based on the separability of the data



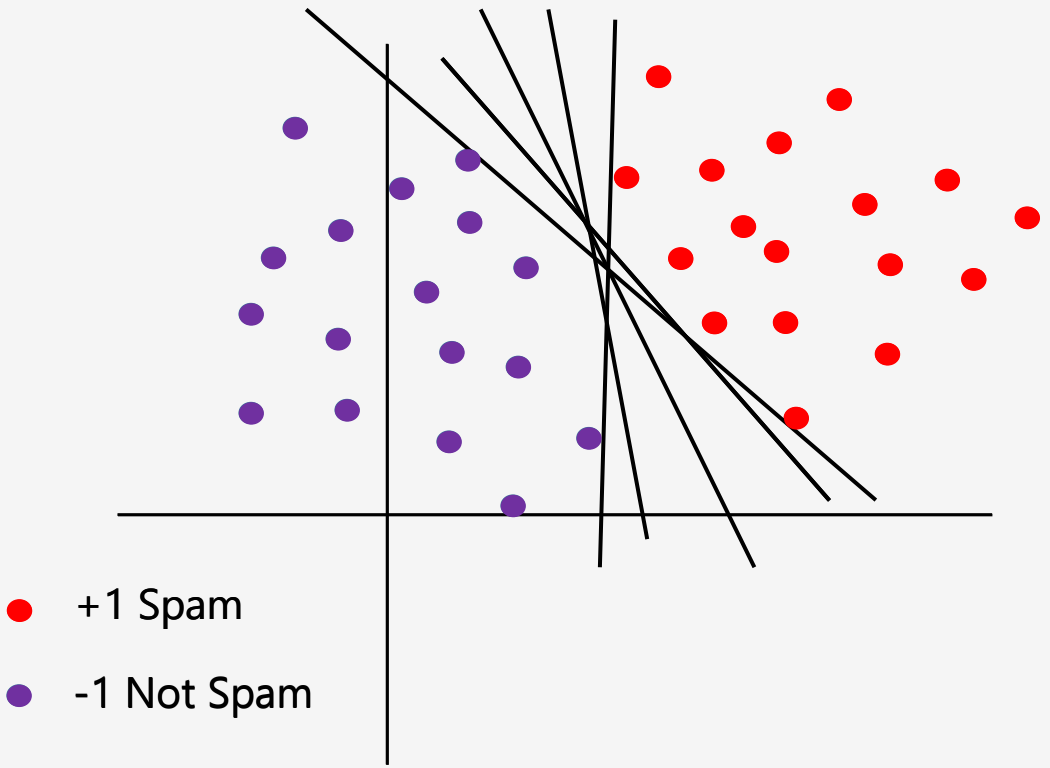
Not linearly separable



Linearly separable

Linearly Separable case

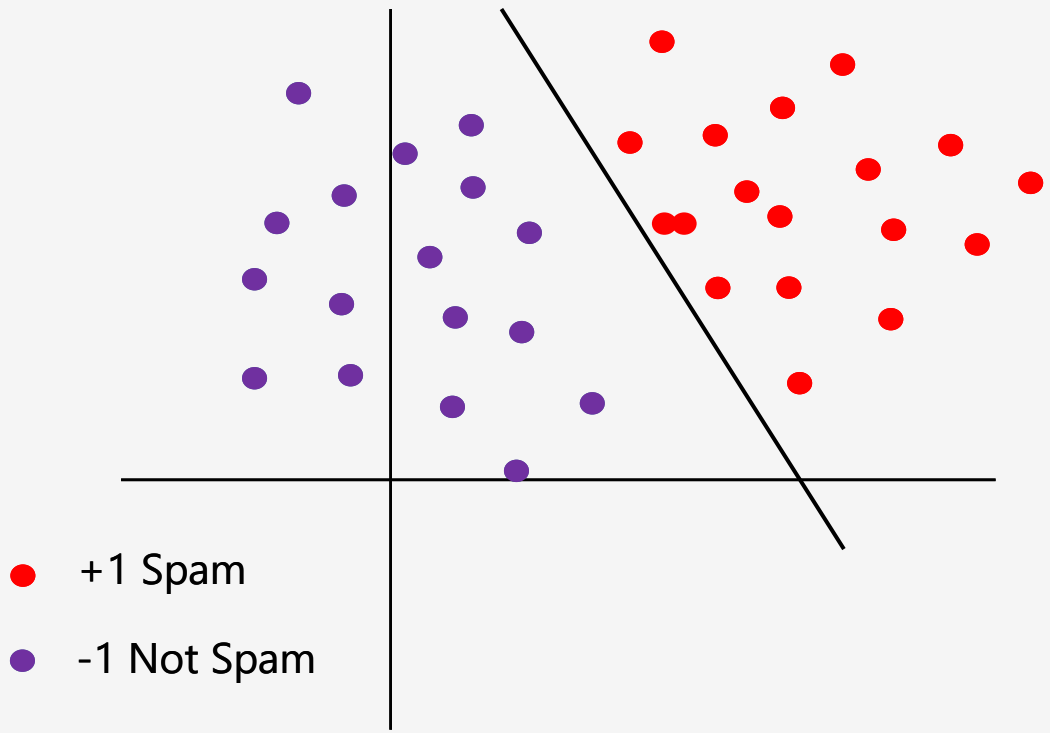
- There are **many affine functions** that separate the data
- **Which one is the best??**
- Your Objective is: find the best separating hyperplane
- Your objective better be convex function
- Any intuition from YOU, which hyperplane is the best??

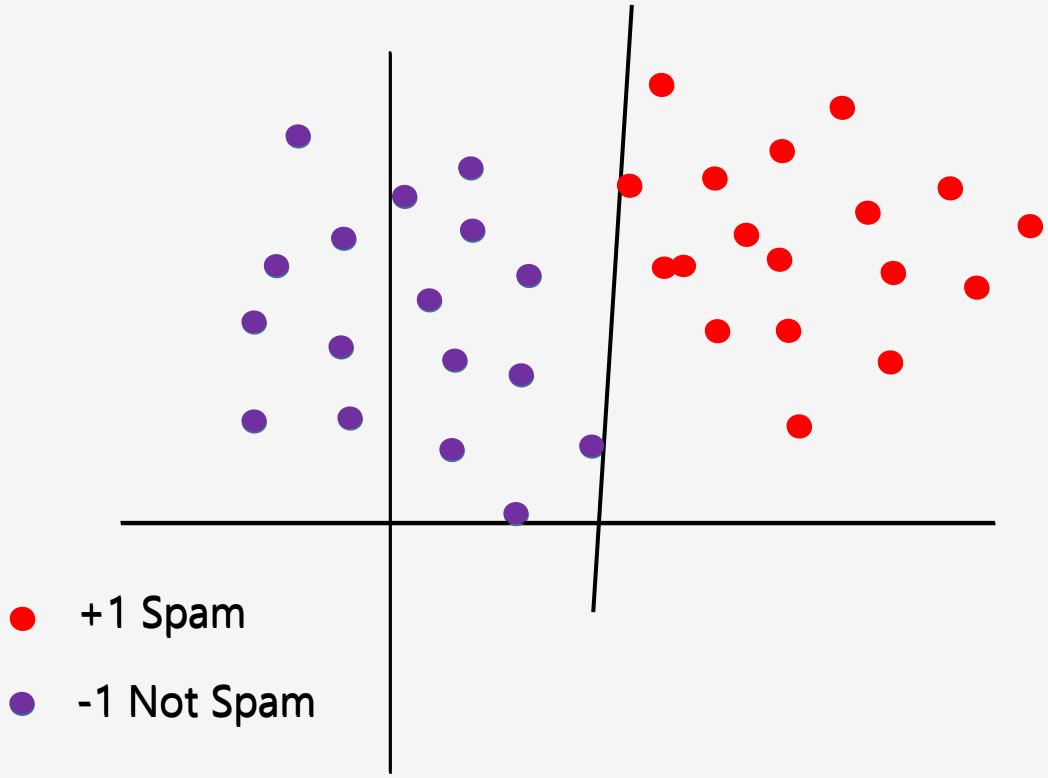


Linearly separable

Linearly Separable case

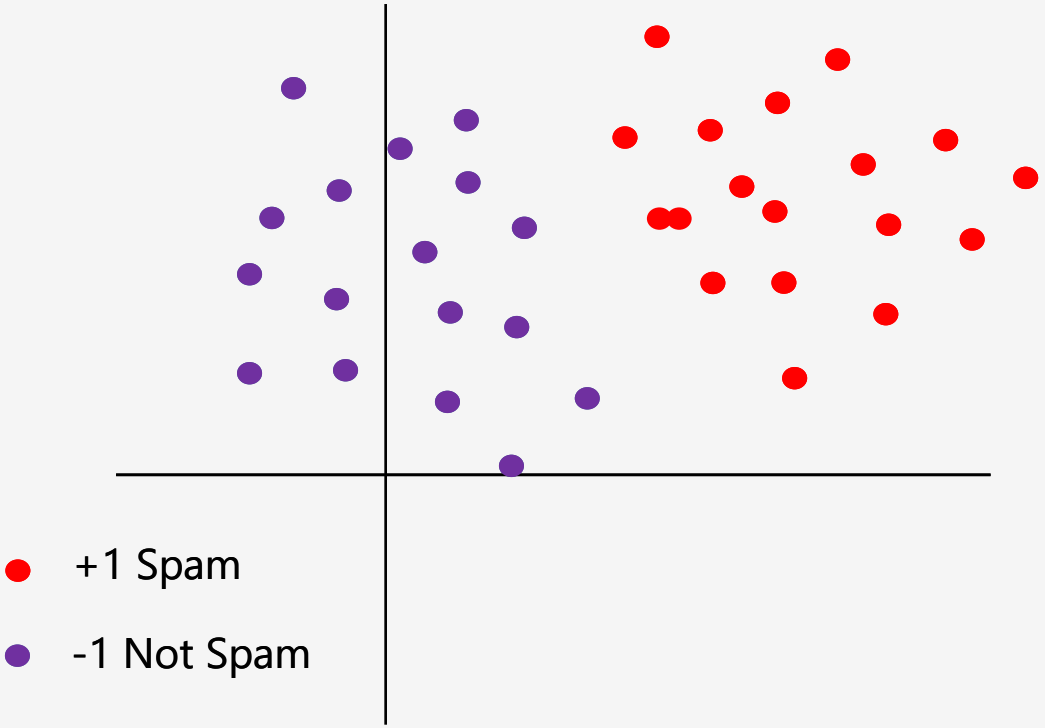
- Let's try some hyperplanes
  - **Tell me which hyperplane is better??**

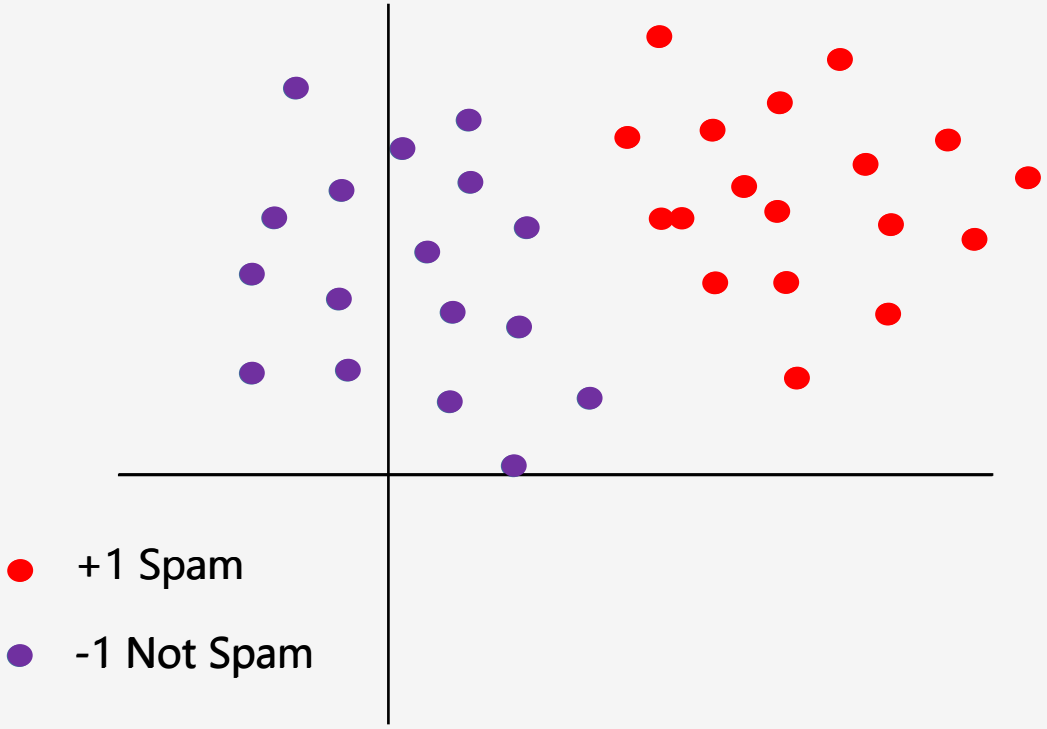




Linearly Separable case

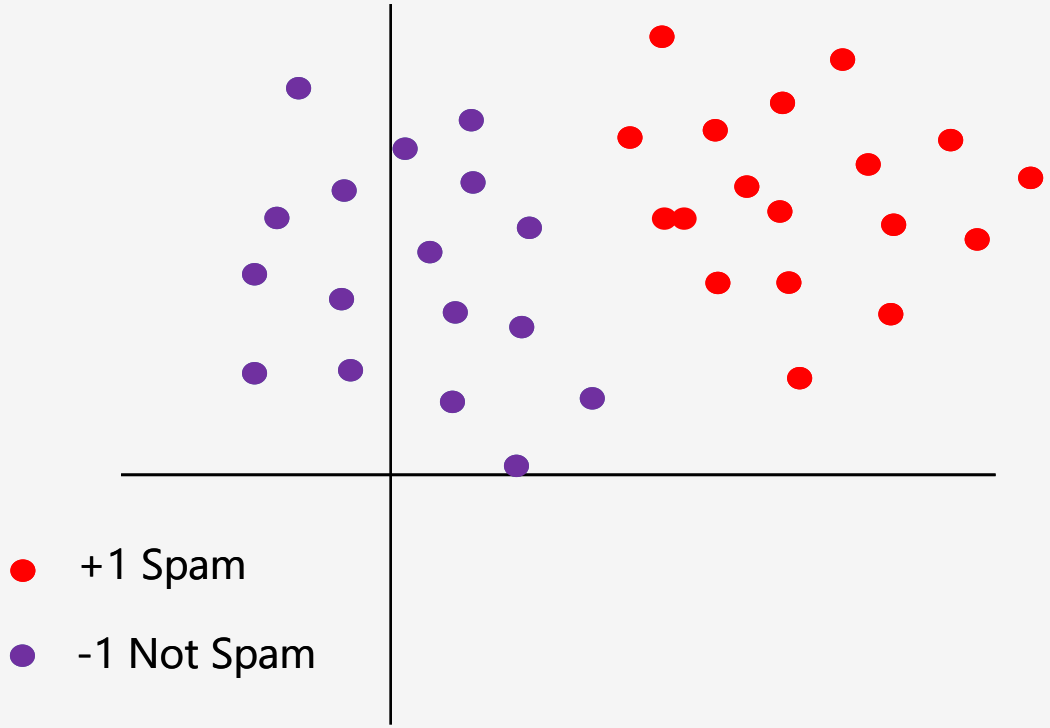
- Any **intuition** about the **best** hyperplane (classifier)??
  - You tell me and I draw…

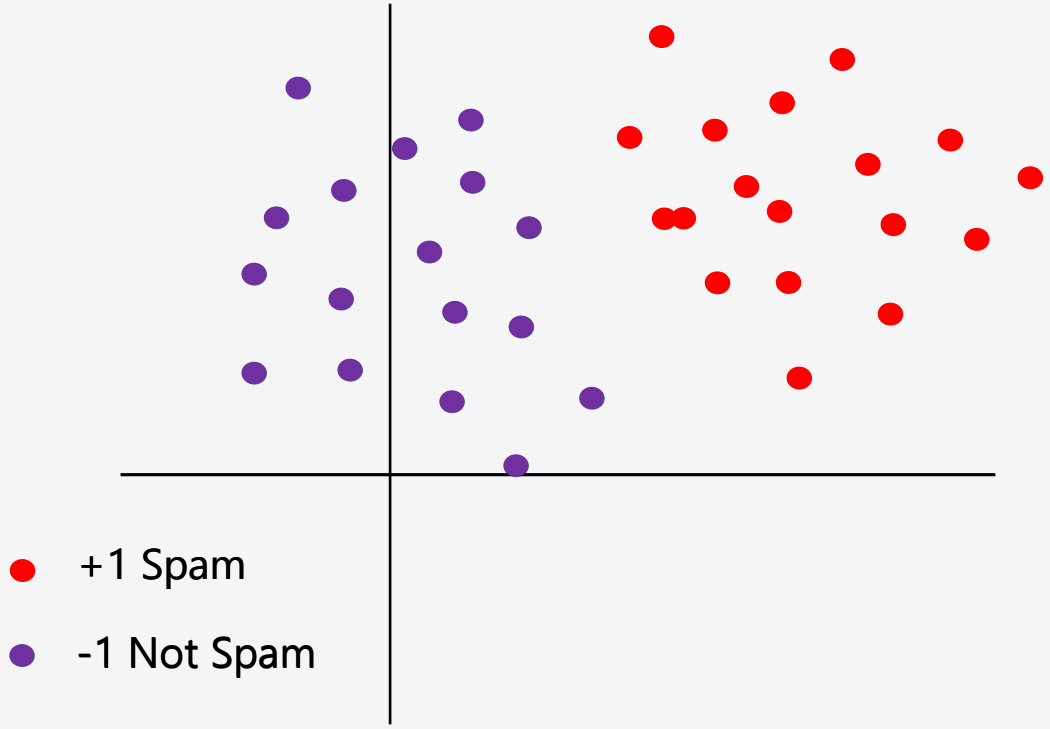




Linearly Separable case

- Any intuition about the best classifier??
- There are infinitely many affine functions that separate the data, **BUT** we want the one that has **lies equally away from the two classes**





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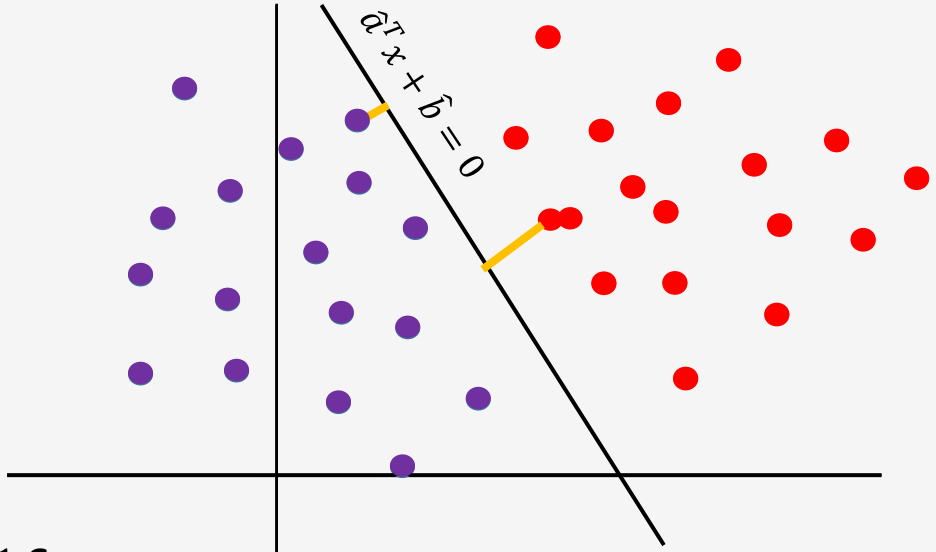
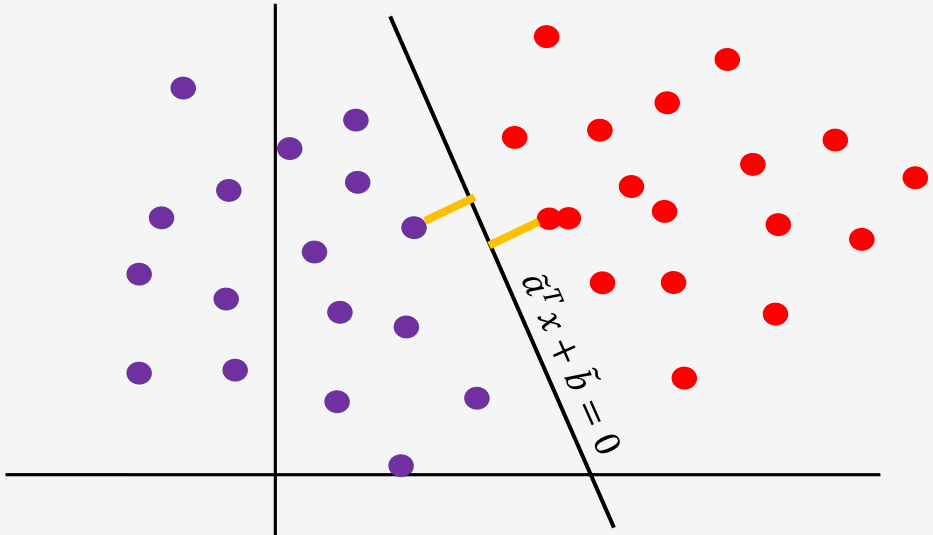


+1 Spam

-1 Not Spam

It is closer to the purple class



+1 Spam

-1 Not Spam

Equally away from both classes

Mathematical definition of the SVM problem

- Recall we are looking for $a \in R^n$ and $b \in R$ such that

$$\begin{aligned} &\text{If} \quad a^T x_i - b \geq 0, \quad \text{then} \quad y_i = +1, \forall i \\ &\text{If} \quad a^T x_i - b \leq 0, \quad \text{then} \quad y_i = -1, \forall i \end{aligned}$$

- Now let's define **mathematically** the SVM problem
  - It is defined as a **constrained optimization problem**

$$\begin{aligned} \min_{a \in R^n, b \in R} \quad & \|a\| \\ s.t. \quad & \left.\begin{aligned} a^T x_i - b \geq +1, \text{if } y_i = +1, \ \forall i \\ a^T x_i - b \leq -1, \text{if } y_i = -1, \ \forall i \end{aligned}\right\} \Leftrightarrow \ y_i \left(a^T x_i - b\right) \geq 1, \forall i \end{aligned}$$

- **Question**: what type of optimization problem is the SVM??

- CVX software can solve very large instances of it!!

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Question: Do you see any difference in the two problems??
Hint: look at the right-hand side of the inequalities

- **Question**: what type of optimization problem is the SVM??

- CVX software can solve very large instances of it!!

Equivalent formulations

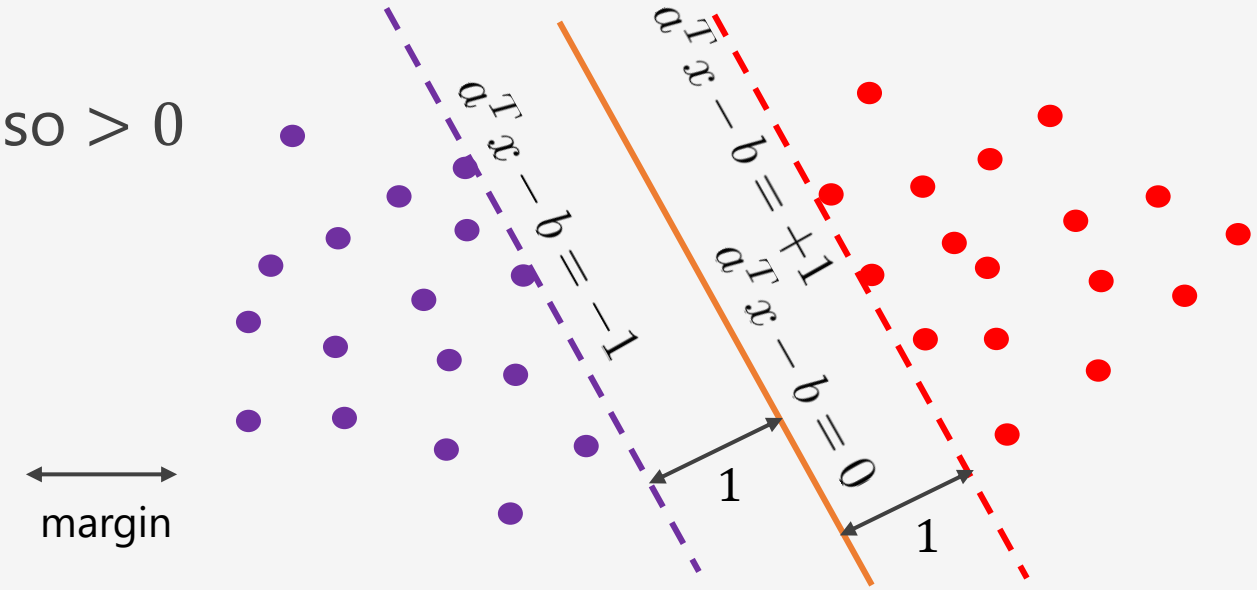
$$(1) \qquad\qquad\qquad\qquad\qquad\qquad\qquad\qquad (2)$$

$$\begin{array}{l} \exists\ a \in R^n, b \in R \\ \text{If} \quad a^T x_i - b \geq \boxed{0,} \quad \text{then} \quad y_i = +1, \forall i \\ \text{If} \quad a^T x_i - b \leq \boxed{0,} \quad \text{then} \quad y_i = -1, \forall i \end{array} \quad \Longleftrightarrow \quad \begin{array}{l} \exists\ a \in R^n, b \in R \\ \text{If} \quad a^T x_i - b \geq \boxed{+1}, \quad \text{then} \quad y_i = +1, \forall i \\ \text{If} \quad a^T x_i - b \leq \boxed{-1}, \quad \text{then} \quad y_i = -1, \forall i \end{array}$$

Both problems are equivalent i.e., they have the same solutions

$\Longleftarrow$ if a solution is ≥ 1, then it is also > 0

$\Longrightarrow$ rescaling



Equivalent formulations

$$\min_{a \in R^n, b \in R} \quad \boxed{\|a\|} \qquad (3)$$
$$s.t. \quad a^T x_i - b \geq +1, \text{if } y_i = +1, \ \forall i$$
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HW4 Prob.1 $\Longleftrightarrow$

$$\max_{a \in R^n, b \in R, t \in R} \quad t \qquad (4)$$
$$s.t. \quad a^T x_i - b \geq +t, \text{if } y_i = +1, \ \forall i$$
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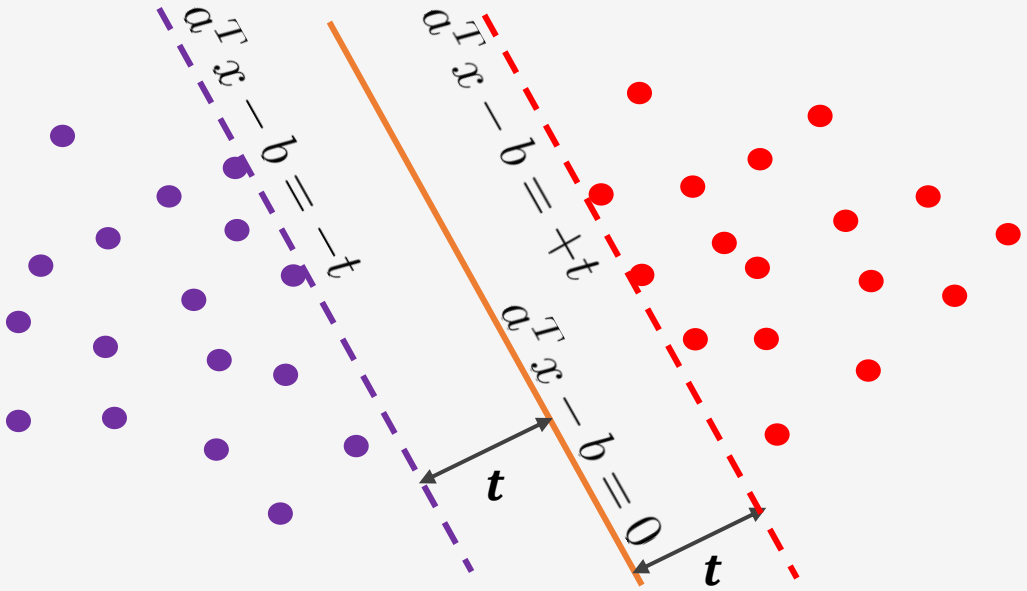
$$\boxed{\|a\| \leq 1}$$

- (3) and (4) are **equivalent**
- problem (4) is easier to **interpret** →

  Maximize the margin t



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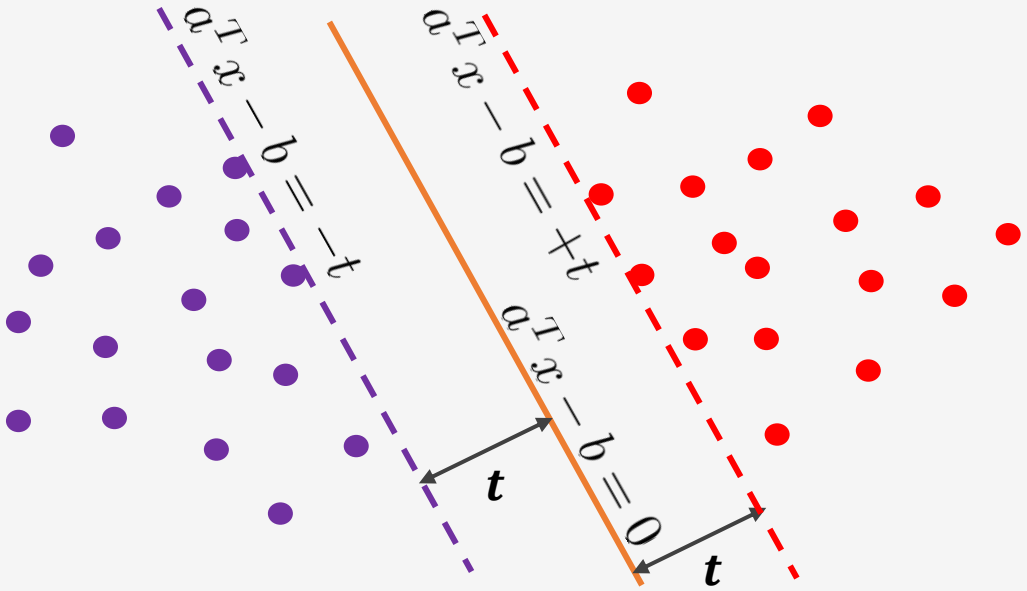
- $||a|| \leq 1$ is needed to normalize solution
- **Why do we need this constraint??**



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Normalization constraint

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  Maximize the margin t

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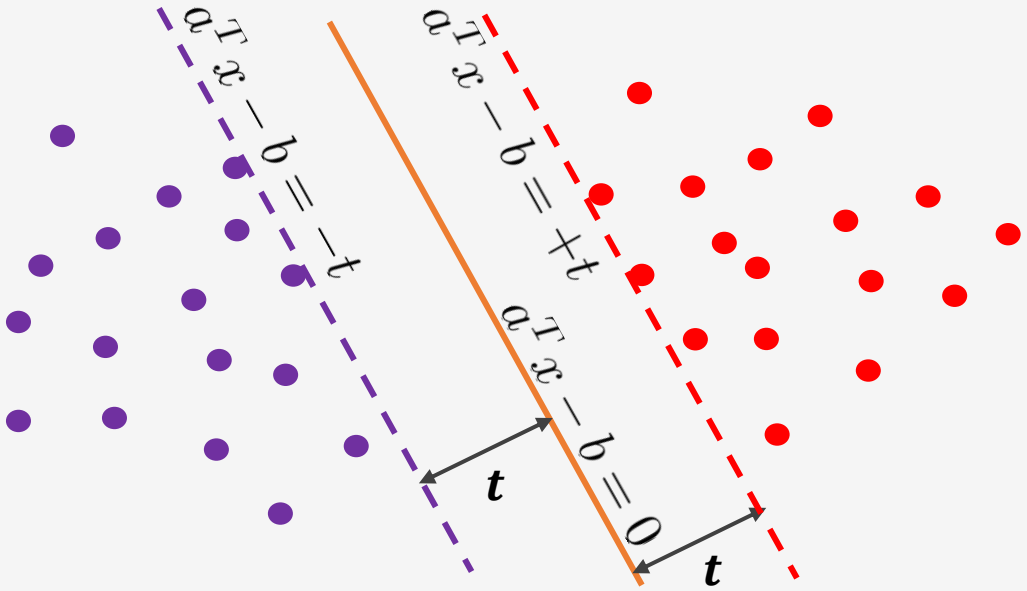
- **Why do we need this constraint??**

  If we did not have it, then $t \to +\infty$



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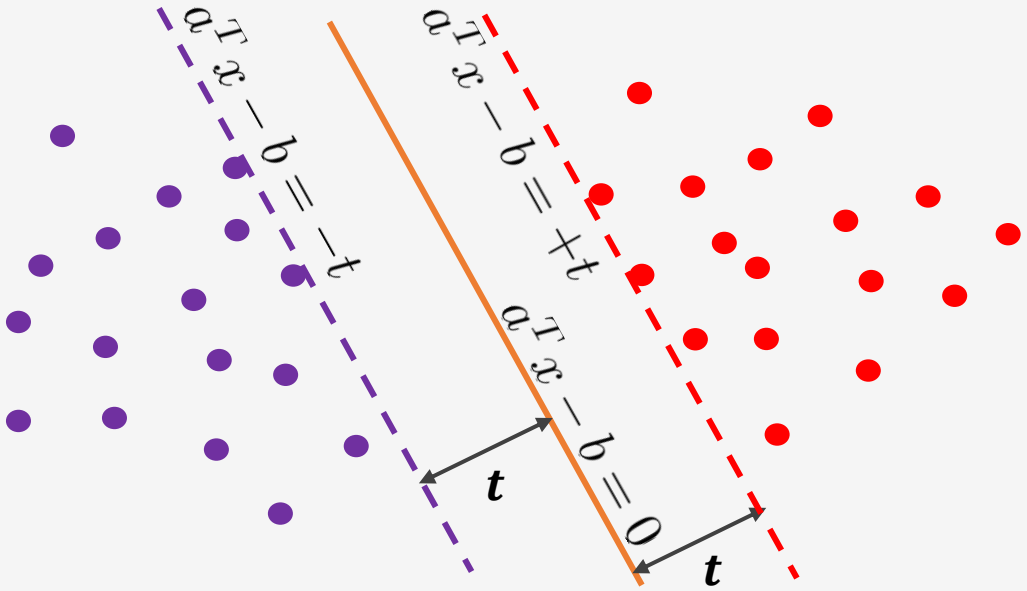

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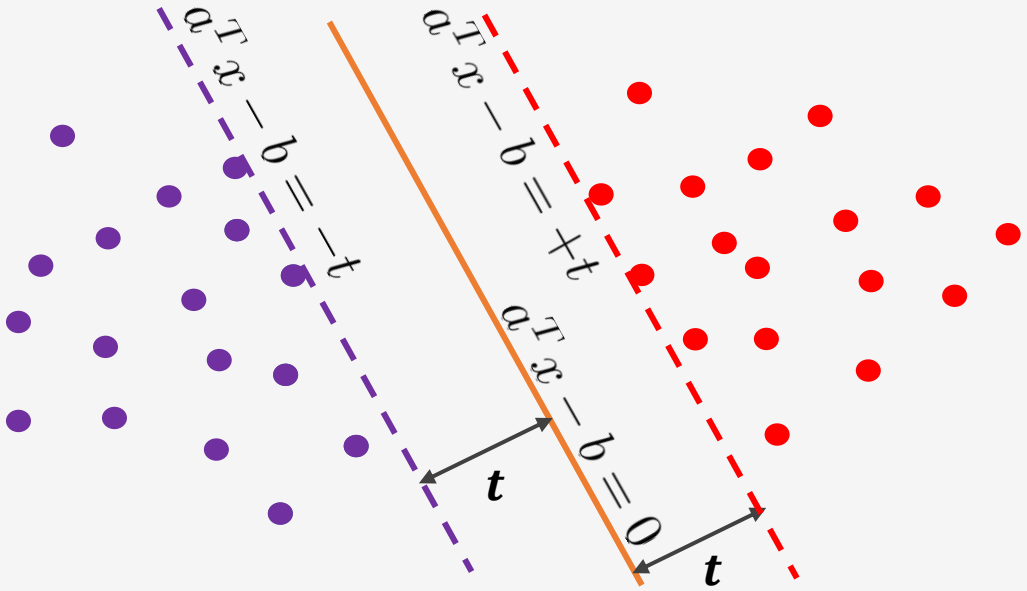

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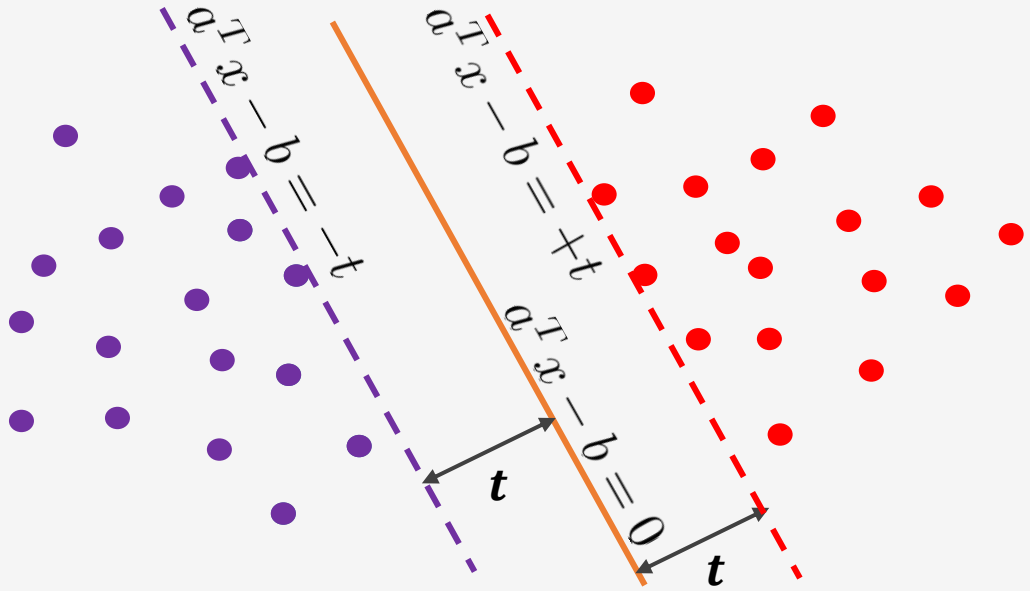
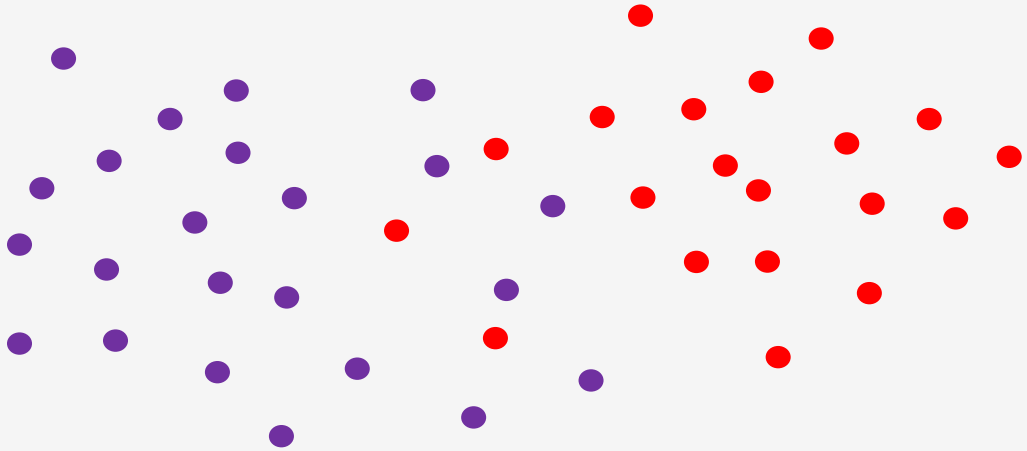
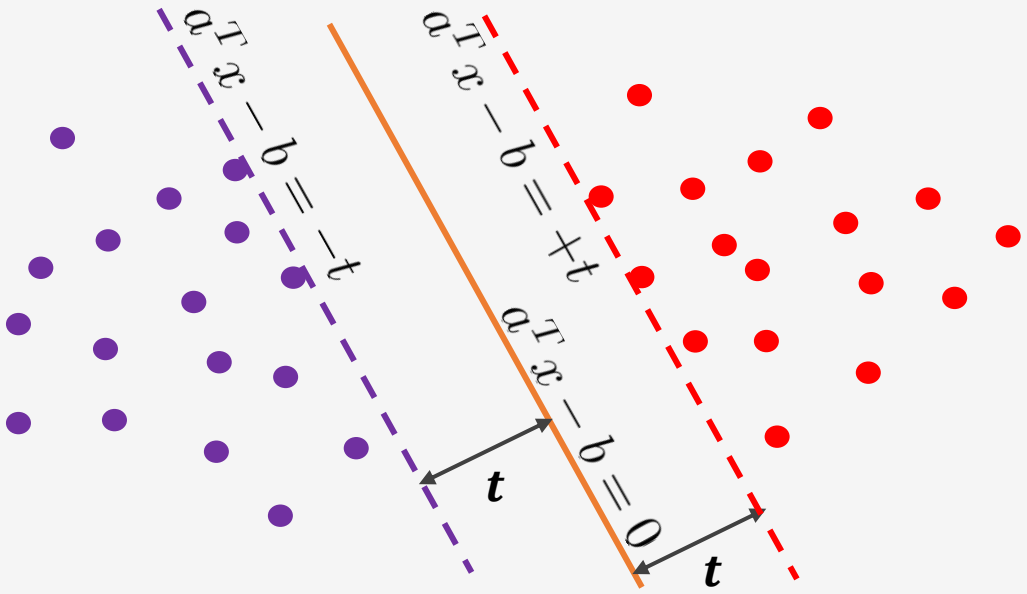


Why did we include $||a|| \leq 1$ in problem (4), instead of $||a|| = 1$??

$||a|| = 1$ would have made problem (4) a <u>nonconvex</u> optimization problem

How to detect if data is linearly separable or not ??





Convex Hulls:

Given $S \subset R^n$ the convex hull of S is

$$convS = \left\{ z = \sum_{i=1}^{n} \lambda_i x_i \mid x_i \in S, \lambda_i \geq 0, \sum_{i=1}^{n} \lambda_i = 1 \right\}$$

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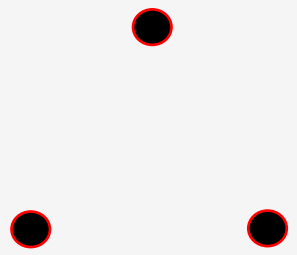
Can we draw some points in it??



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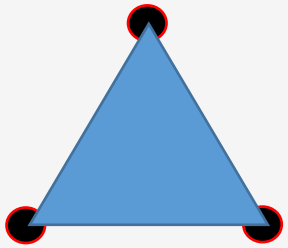

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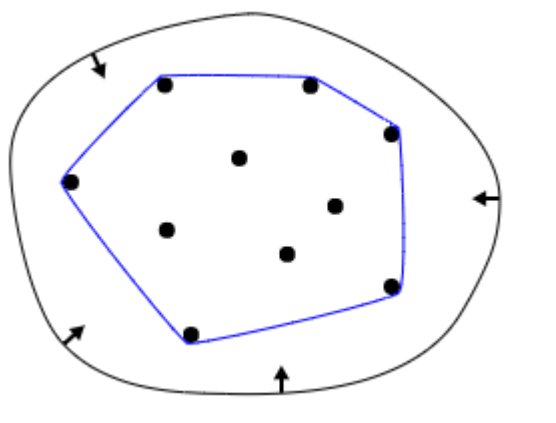
S is a set of points in R^n

The convex hull of S is the <u>smallest</u> convex set that contains S



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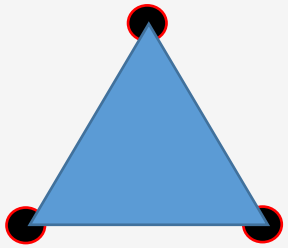

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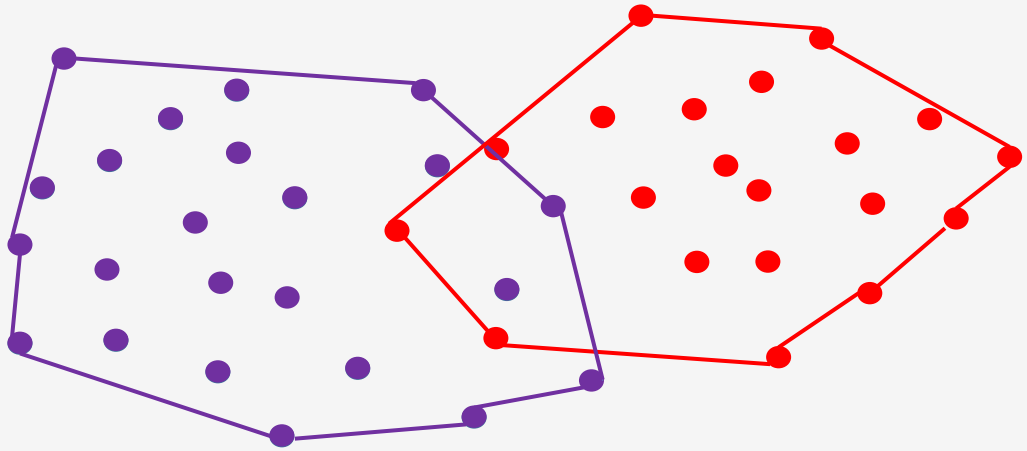
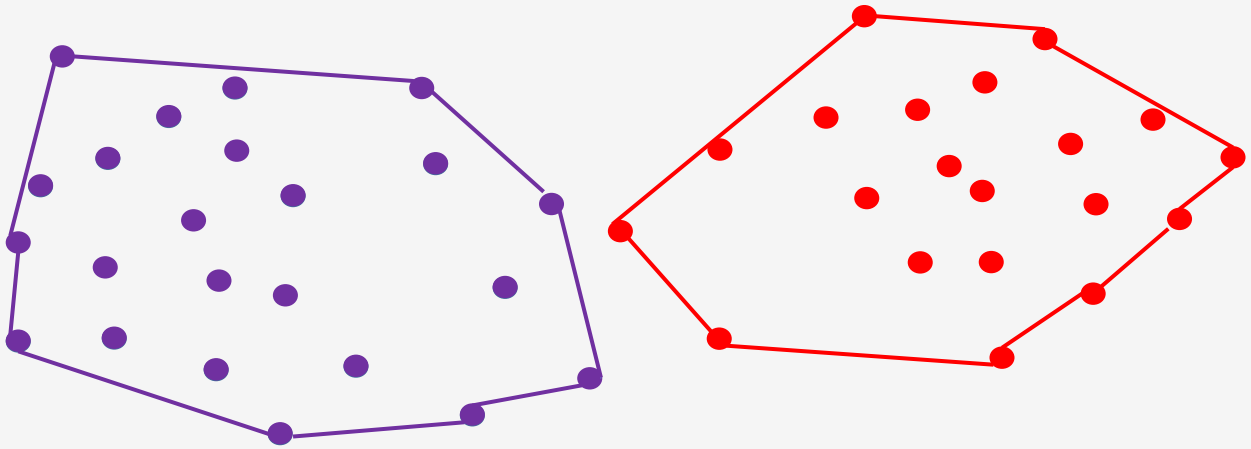
How to detect if data is linearly separable??





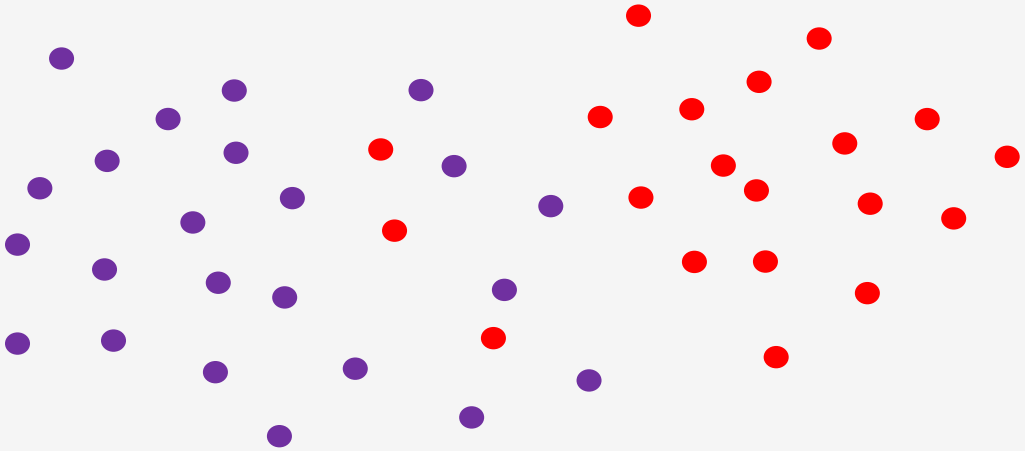
Question:

What is the connection between convex hulls and non-separable datasets??

The **intersection** of the convex hulls of **non-separable** datasets is **non-empty**

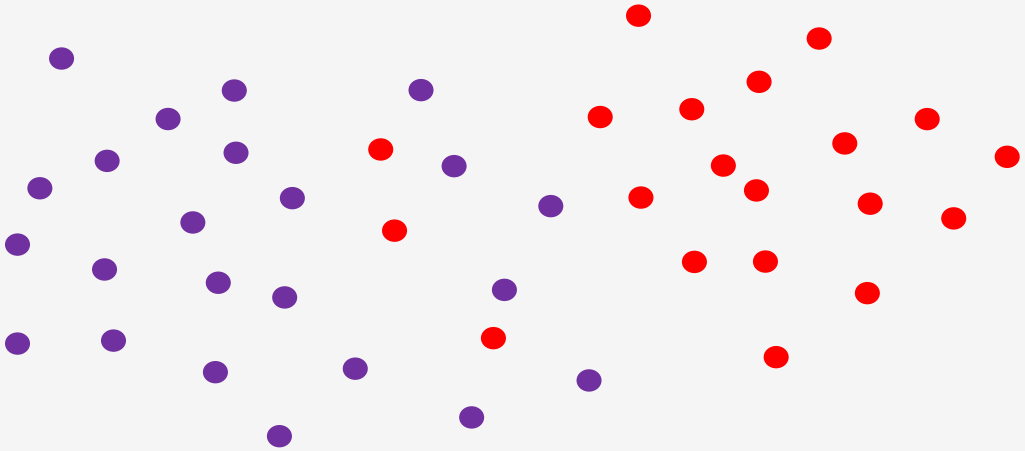
The **intersection** of the convex hulls of **separable** datasets is **empty**

SVMs when data is not linearly separable



Question: What is a good hyperplane for non-separable data??

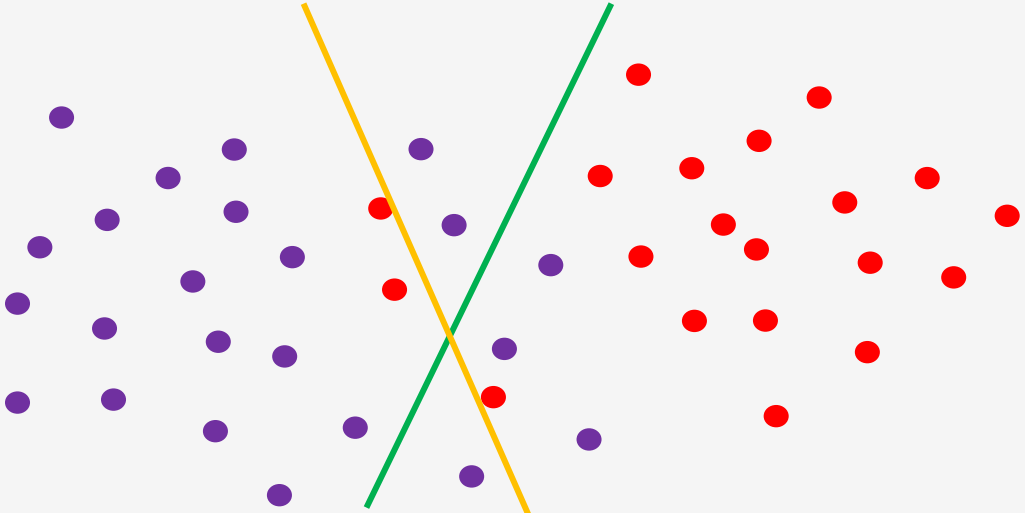
SVMs when data is not linearly separable



Question: What is a good hyperplane for non-separable data??

A hyperplane that **minimizes the number of misclassified data points in the training set**

SVMs when data is not linearly separable



Question: which line is better classifier??

Question: What is a good hyperplane for non-separable data??

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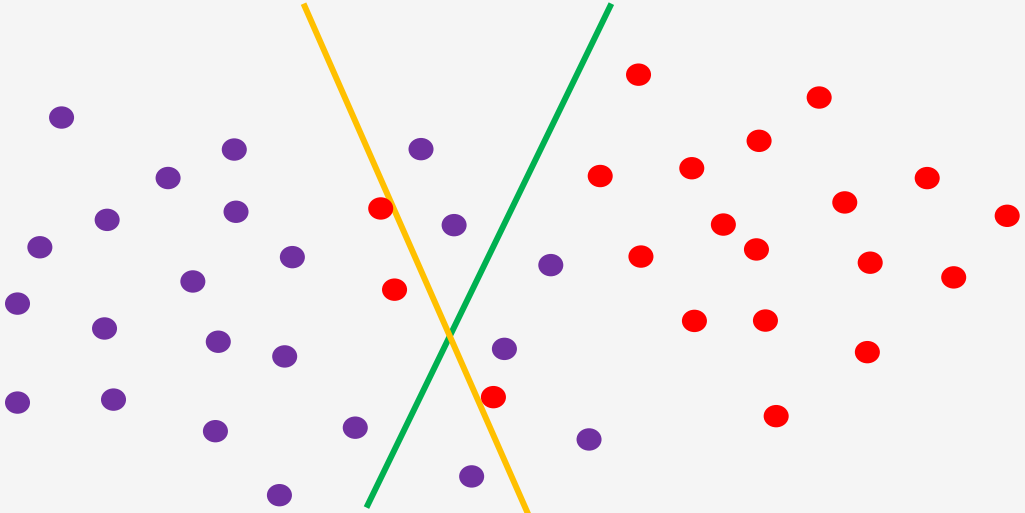
SVMs when data is not linearly separable



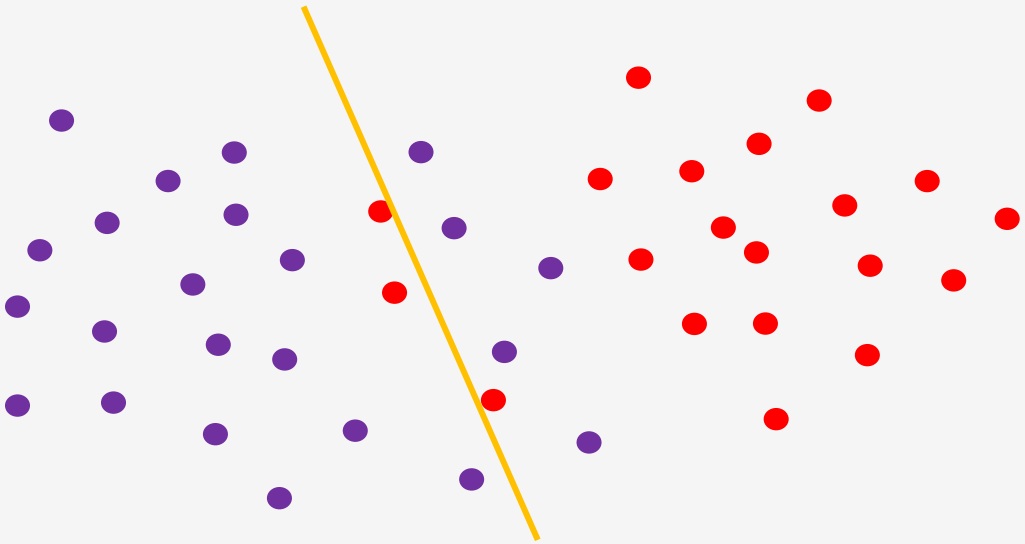
Question: which line is better classifier??

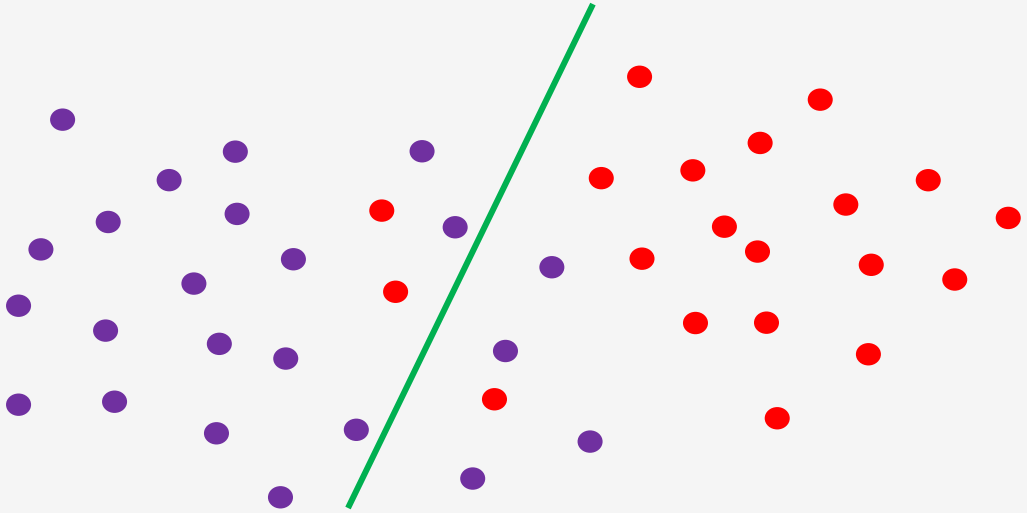
Hint: We are looking for the number of misclassified ●

Question: What is a good hyperplane for non-separable data??

A hyperplane that **minimizes the number of misclassified data points in the training set**

SVMs when data is not linearly separable





A **hyperplane** that **minimizes the number of misclassified data points** in the training set

$$\min_{a \in R^n, b \in R, \eta \in R^m} \quad \boxed{\|\eta\|_0}$$

$$\begin{aligned} s.t. \quad & a^T x_i - b \geq \boxed{+1 - \eta_i}, \ \text{if } y_i = +1, \ \forall i \\ & a^T x_i - b \leq \boxed{-1 + \eta_i}, \ \text{if } y_i = -1, \ \forall i \\ & \eta_i \geq 0, \ \forall i \end{aligned}$$

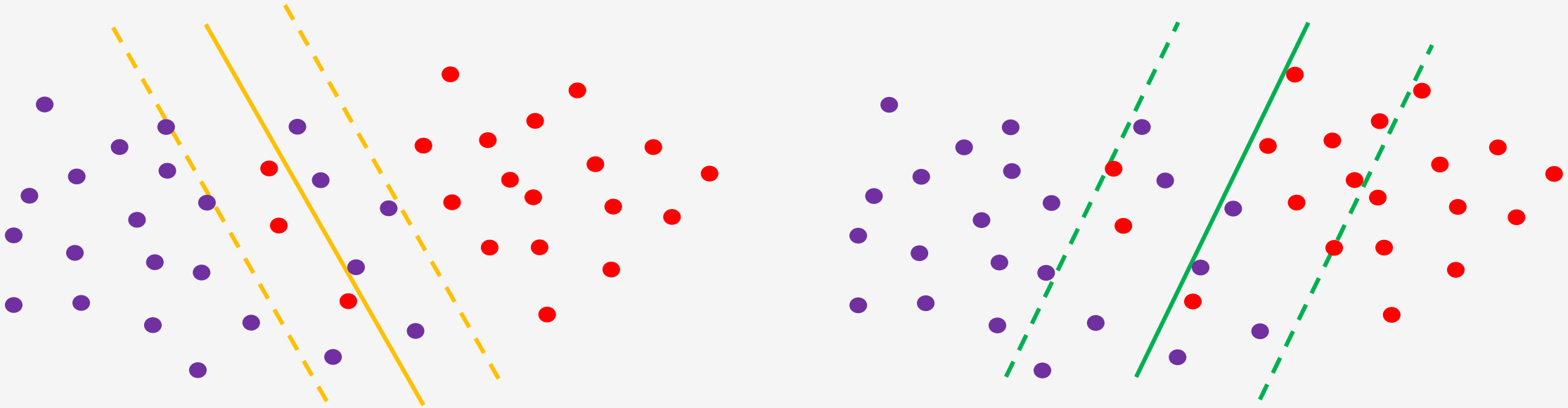
Relaxed constraints

$\|\eta\|_0$ is the L_0 norm
Represents the # of nonzeros in η.

$$\eta = (n_1, \eta_2, \ldots, \eta_m)^T$$

Non-convex optimization problem →
$||\eta||_0$ is a nonconvex function

SVMs when data is not linearly separable



A **hyperplane** that **minimizes the number of misclassified data points** in the training set

$$\min_{a \in R^n, b \in R, \eta \in R^m} \boxed{\|\eta\|_1}$$

Relaxed constraints

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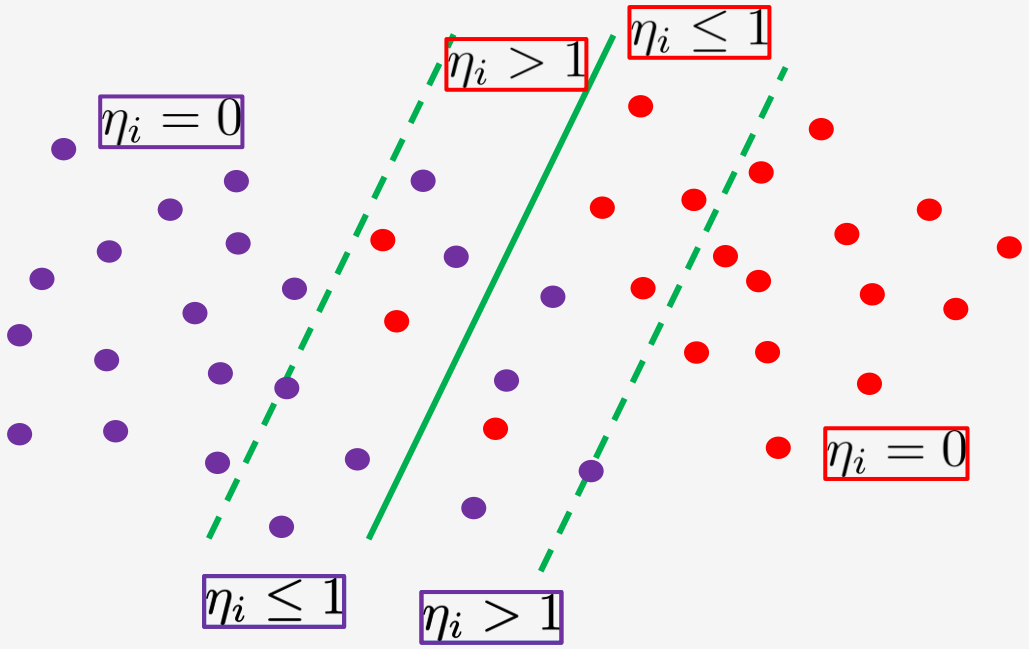
Convex optimization problem → $||\eta||_1$ is a convex function



If $\eta_i = 0$ or $\eta_i \leq 1$ then x_i is correctly classified (for both ● and ●)

If $\eta_i > 1$ then x_i is incorrectly classified (for both ● and ●)

Soft margin extension of SVMs

Multi-objective optimization problem

minimize # of misclassified points

maximize the margin

$$
\begin{aligned}
\min_{a \in R^n, b \in R, \eta \in R^m} \quad & \|\eta\|_1 + \gamma \|a\| \\
s.t. \quad & a^T x_i - b \geq +1 - \eta_i, \ \text{if } y_i = +1, \ \forall i \\
& a^T x_i - b \leq -1 + \eta_i, \ \text{if } y_i = -1, \ \forall i \\
& \eta_i \geq 0, \ \forall i
\end{aligned}
$$